Supplementary materials for Sharp tail bounds for the extreme singular values of sub-Gaussian matrices

Marcus Webb
University of Manchester

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Abstract

This entire document, including its mathematical proof and exposition, was generated by artificial intelligence without direct input from the author. Marcus Webb nevertheless takes full responsibility for its content. The accompanying paper proves singular-value tail bounds with unspecified universal constants. Here we give the detailed estimates and finite arithmetic certificates that yield explicit values, with sharper values when $K = 1$. The proof uses the paper's barrier construction and has hybrid verification: Lean checks the analytic argument conditional on finite rational certificates, checked separately by exact Python arithmetic.

S1 Explicit constants: an AI-generated proof

The accompanying paper by Marcus Webb and Morgan Lynch-Davies is available on arXiv. It presents the probabilistic setting and motivation, comparisons with earlier bounds, applications to randomized numerical linear algebra, and sharpness examples and discussion of the correction terms. Its proof explains the barrier method and establishes the tail bounds with unspecified universal constants. The paper also states both numerical quartets established below.

This supplement gives the detailed derivation of those quartets, including the $K = 1$ refinement, in Section S1; Section S2 documents the Lean/Python verification and AI assistance. We refine the proof in Sections 3–5 by estimating the mean cost and fluctuations of the row shifts separately. The theorem statement below is self-contained; its proof uses the paper's deterministic barrier construction. The constants are not claimed to be optimal.

Theorem S1.1 (Explicit singular-value tail bounds)**.** *There are positive universal constants $\alpha, \beta, \gamma, \delta$ with the following property. Let $\mathbf{S} \in \mathbb{R}^{m\times n}$, $m \geq n \geq 1$, have independent, centred, isotropic rows X_i^T, meaning $\mathbb{E}X_i = 0$ and $\mathbb{E}X_iX_i^T = I_n$. Suppose that, for some $K \geq 1$,*

$$\mathbb{E}e^{\lambda\langle X_i, v\rangle} \leq e^{K^2\lambda^2\|v\|_2^2/2} \qquad (\lambda \in \mathbb{R},\ v \in \mathbb{R}^n,\ 1 \leq i \leq m).$$

Put $L = \log(eK)$, with natural logarithms, and let $\sigma_{\min}(\mathbf{S})$ and $\sigma_{\max}(\mathbf{S})$ denote the smallest and largest of its n singular values, including zeros. Then, for every $t \geq 0$,

$$\mathbb{P}\left\{\sigma_{\min}(\mathbf{S}) \leq \sqrt{m} - K\sqrt{n} - \alpha KL - \beta K\sqrt{L}\,t\right\} \leq e^{-t^2/2}, \tag{S1.1}$$

$$\mathbb{P}\left\{\sigma_{\max}(\mathbf{S}) \geq \sqrt{m} + K\sqrt{n} + \gamma KL + \delta K\sqrt{L}\,t\right\} \leq e^{-t^2/2}. \tag{S1.2}$$

For every $K \geq 1$, one may take

$$(\alpha, \beta, \gamma, \delta) = (3.84, 1.58, 2.35, 1.5). \tag{S1.3}$$

When $K = 1$, one may instead take

$$(\alpha, \beta, \gamma, \delta) = (1.32, 1.12, 2.32, \sqrt{2}). \tag{S1.4}$$

Guide to the proof. The coefficients α, γ measure the additive correction KL; β, δ measure the fluctuation scale $K\sqrt{L}\,t$. The proof separates these two costs:

1. Section S1.1 converts one-row Laplace or MGF estimates into singular-value tails, keeping the correction and fluctuation costs separate.
2. Sections S1.2 and S1.3 collect the common probabilistic tools and the exact arithmetic rules used in the certificates.
3. Sections S1.4–S1.5 prove the general lower pair (α, β); Section S1.6 sharpens it at $K = 1$.
4. Section S1.7 proves both upper pairs (γ, δ), first controlling the mean and then the exponential moment of the upper shift.

The scalar certificates are included where they enter the proof, so that each numerical bound can be traced to a finite check. The implementation and its correspondence with the Lean hypotheses are described in Sections S2.3–S2.4.

Notation and barrier shifts. For symmetric matrices, $R \succ 0$ and $R \succeq 0$ mean positive definite and positive semidefinite, respectively. We write $\|R\|$ for the operator norm and $\|R\|_F = (\operatorname{tr}(R^2))^{1/2}$ for the Frobenius norm. The matrix R is the current barrier resolvent. The nonnegative exact shifts Δ_- and Δ_+ preserve the trace of the resolvent after a row update; their defining equations are recalled in Sections S1.4 and S1.7. In a one-row estimate R is deterministic; in the matrix construction it is determined by the preceding rows. The theorem's parameter t is reserved throughout for the final tail bound.

S1.1 From one-row estimates to the four constants

Lemma S1.2 (Separate drift and fluctuation costs)**.** *Fix $K \geq 1$. Suppose $w, \nu > 0$ have one of the following properties for every deterministic $R \succ 0$ and every row X satisfying the assumptions of Theorem S1.1.*

1. *Whenever $K^2 \operatorname{tr} R + w\|R\|_F \leq b$, $b > 0$, the exact lower shift satisfies*

$$\mathbb{E}e^{-\lambda(1+b)\Delta_-} \leq \exp\left(-\frac{\lambda}{1+\nu\lambda}\right) \qquad (\lambda > 0).$$

   Then the lower tail of the theorem holds with correction w/K and fluctuation term $\sqrt{\nu/2}\,t$.

2. *Whenever $K^2 \operatorname{tr} R + w\|R\|_F \leq b < 1$, the exact upper shift satisfies*

$$\mathbb{E}e^{\lambda(1-b)\Delta_+} \leq \exp\left(\frac{\lambda}{1-\nu\lambda}\right) \qquad (0 < \lambda < 1/\nu).$$

   Then the upper tail of the theorem holds with correction w/K and fluctuation term $\sqrt{\nu/2}\,t$.

Here w controls the correction cost and ν controls the fluctuations. In particular, $w = AK^2L$ and $\nu = BK^2L$ give the correction coefficient A and fluctuation coefficient $\sqrt{B/2}$.

Proof. Use the same trace-preserving row shifts and outward corrections as in Section 4, now imposing $K^2 \operatorname{tr} R + w\|R\|_F \leq b$. Write $S_\pm$ for the sums of the m adaptive row shifts $\Delta_{\pm,i}$ in the two constructions. Starting at distance $(K^2 n + w\sqrt{n})/b$, the trace integral gives

$$\sum_i \omega_i \leq \frac{w\sqrt{n}}{b}, \qquad d_w = \sqrt{K^2 n + 2w\sqrt{n}} < D_w := K\sqrt{n} + \frac{w}{K}.$$

Thus the squared singular values are bounded by the sums of row shifts, with total initial and outward cost at most d_w^2/b. This is the same calculation as (4.1), with w in place of v in the regularity condition only.

Take $t > 0$ and put $s_\pm = \sqrt{m} \pm \sqrt{\nu/2}\, t$. For the lower tail, the target $s_- - D_w < 0$ gives an empty event. Otherwise $s_- > d_w > 0$, so choose $b = d_w/(s_- - d_w)$ and $\lambda = (\sqrt{m} - s_-)/(\nu s_-) > 0$. Condition on preceding rows and iterate the lower one-row estimate. This standard exponential iteration is also used in [3, Theorem 2.6]. Markov's inequality gives

$$\mathbb{P}\{(1+b)S_- \le s_-^2\} \le \exp\!\left(\lambda s_-^2 - \frac{m\lambda}{1+\nu\lambda}\right) = \exp\!\left(-\frac{(\sqrt{m} - s_-)^2}{\nu}\right) = e^{-t^2/2}.$$

Outside this event,

$$\sigma_{\min}(\mathbf{S})^2 > \frac{s_-^2}{1+b} - \frac{d_w^2}{b} = (s_- - d_w)^2 > (s_- - D_w)^2.$$

This includes a zero target because $D_w > d_w$.

For the upper tail take $b = d_w/(s_+ + d_w) < 1$ and $\lambda = (s_+ - \sqrt{m})/(\nu s_+)$, so $0 < \lambda < 1/\nu$. The corresponding iteration and Markov inequality give

$$\mathbb{P}\{(1-b)S_+ \ge s_+^2\} \le \exp\!\left(-\lambda s_+^2 + \frac{m\lambda}{1-\nu\lambda}\right) = \exp\!\left(-\frac{(s_+ - \sqrt{m})^2}{\nu}\right) = e^{-t^2/2}.$$

Outside this event,

$$\sigma_{\max}(\mathbf{S})^2 < \frac{s_+^2}{1-b} + \frac{d_w^2}{b} = (s_+ + d_w)^2 < (s_+ + D_w)^2.$$

Taking square roots proves both claims; $t = 0$ is immediate. □

S1.2 Gaussian comparison and weighted quadratic forms

The following form of the Hsu–Kakade–Zhang comparison keeps the constants visible. It is [4, Remark 2.3]; we include the short deduction for the normalization used here.

Lemma S1.3 (Gaussian comparison)**.** *For $B \succeq 0$ and $\lambda \ge 0$ such that $I - 2K^2\lambda B \succ 0$,*

$$\mathbb{E}e^{\lambda X^T B X} \le \det(I - 2K^2\lambda B)^{-1/2}.$$

Proof. For a standard Gaussian vector g independent of X, the row MGF assumption gives

$$\mathbb{E}e^{\lambda X^T B X} = \mathbb{E}_g \mathbb{E}_X e^{\sqrt{2\lambda}\langle B^{1/2} g, X\rangle} \le \mathbb{E}_g e^{K^2 \lambda g^T B g} = \det(I - 2K^2\lambda B)^{-1/2}.$$

□

Lemma S1.4 (Weighted entropy inequality)**.** *Let $W \ge 0$ have $\mathbb{E}W = 1$ and finite entropy $D(W) = \mathbb{E}(W \log W)$, where $0 \log 0 = 0$. Every random variable Z with $\mathbb{E}e^Z < \infty$ satisfies*

$$\mathbb{E}(WZ) \le D(W) + \log \mathbb{E}e^Z, \tag{S1.5}$$

whenever its left side is defined. No independence between W and Z is required.

Proof. Use the probability measure $W\, d\mathbb{P}$. Apply concavity of the logarithm to e^Z/W on $\{W > 0\}$, first using bounded truncations if necessary; the expectation under $W\, d\mathbb{P}$ is at most $\mathbb{E}e^Z$. This proves (S1.5). □

S1.3 Exact arithmetic certificates

The numerical estimates below use rational enclosures rather than floating-point approximations. We use two rules: Taylor series with explicit remainders bound exponentials and logarithms, and Bernstein coefficients certify polynomial inequalities on whole intervals or rectangles.

S1.3.1 Exponential and logarithmic enclosures

For integers $N \geq 0$, put $T_N(x) = \sum_{j=0}^{N} x^j/j!$. Then $T_N(x) \leq e^x$ for every $x \geq 0$, and

$$T_N(x) \leq e^x \leq T_N(x) + \frac{x^{N+1}}{(N+1)!}\frac{1}{1-x/(N+2)} \qquad (0 \leq x < N+2). \tag{S1.6}$$

For integers $N \geq 1$, put $S_N(z) = 2\sum_{j=0}^{N-1} z^{2j+1}/(2j+1)$. Then

$$S_N(z) \leq \log\frac{1+z}{1-z} \leq S_N(z) + \frac{2z^{2N+1}}{(2N+1)(1-z^2)} \qquad (0 \leq z < 1). \tag{S1.7}$$

Both remainders are bounded by geometric series. For negative exponentials take reciprocals; for logarithms use $z = (x-1)/(x+1)$, first reducing x by powers of two when specified. For $0 < x < 1$, use $\log x = -\log(1/x)$. The certificates state their truncation orders and precision: lower bounds are rounded down and upper bounds up, with each substitution chosen according to the sign of its multiplier.

S1.3.2 Bernstein coefficients

Let $p(x) = \sum_{r=0}^{d} c_r x^r$ and $h > 0$. Put

$$e_j = \sum_{r=j}^{d} c_r \binom{r}{j} A^{r-j} h^j, \qquad b_i = \sum_{j=0}^{i} e_j \frac{\binom{i}{j}}{\binom{d}{j}}.$$

Then

$$p(A+hu) = \sum_{i=0}^{d} b_i \binom{d}{i} u^i (1-u)^{d-i}.$$

The identity follows by expanding both sides in powers of u. Its basis functions are nonnegative and sum to one for $0 \leq u \leq 1$, so $p(x) \geq \min_i b_i$ on $[A, A+h]$. Applying this rule in each coordinate gives the same property for tensor-product Bernstein coefficients on a rectangle.

S1.4 The lower-tail constants

Lower-barrier criteria appear in Batson, Spielman and Srivastava [2, Lemma 3.4], Kolla et al. [6, Lemma 3.11], and Yaskov [8, Lemmas 4.1–4.2] and [9, Lemma 4.1]. We use the exact trace-preserving shift and keep its mixed quadratic moment explicit. The proof has three parts: a comparison for one row, an estimate of its cost in terms of the spectrum, and a bound on the total cost of restoring the regularity cap.

S1.4.1 Fourth-moment bounds

Suppose Z is centred, $\mathbb{E}Z^2 = 1$, and $\mathbb{E}e^{uZ} \leq e^{K^2u^2/2}$ for every real u. Write $L = \log(eK)$ and $m_4 = \mathbb{E}Z^4$. Moment interpolation gives $\mathbb{E}Z^{2j} \geq m_4^{j-1}$ for every integer $j \geq 1$, and therefore

$$e^{K^2u^2/2} \geq \mathbb{E}\cosh(uZ) \geq 1 + \frac{\cosh(u\sqrt{m_4}) - 1}{m_4} \geq \frac{e^{u\sqrt{m_4}}}{2m_4}.$$

Taking $u = \sqrt{m_4}/K^2$ gives $m_4/(2K^2) \le \log(2m_4)$. The difference $x/(2K^2) - \log(2x)$ increases for $x > 2K^2$. At $x = 5K^2L$ it is $L/2 + 2 - \log(10L) \ge 3 - \log 20 > 0$, so

$$\mathbb{E}Z^4 \le 5K^2L. \tag{S1.8}$$

We also need the slightly different bound

$$\mathbb{E}Z^4 \le \frac{499}{100}K^2L + \frac{1}{2}. \tag{S1.9}$$

Put $c = 499/100$, $a = c/2 - 2 = 99/200$ and $E = e^{2-2L}$. At $x = cK^2L + 1/2$, the same increasing difference is at least

$$aL + 2 - \log(2cL) + E\left(\frac{1}{4} - \frac{1}{2cL}\right) > aL + 2 - \log(2cL) + \frac{7}{50}E.$$

For $1 \le L \le 5/2$, this is greater than $3 - \log(1996/99) + (7/50)e^{-3} > -1/250 + 1/150 > 0$. For $L \ge 5/2$, the first three terms increase and are already positive at $5/2$, since $\log(499/20) < 259/80$. The numerical comparisons follow from degree-24 Taylor bounds: $e^{751/250} > 1996/99$, $e^3 < 21$, and $e^{259/80} > 499/20$. This proves (S1.9).

For $P \succeq 0$, $\operatorname{tr} P = 1$, Gaussian averaging gives

$$T = X^TPX: \quad \mathbb{E}T = 1, \qquad \mathbb{E}T^2 \le \frac{5}{3}K^2L[1 + 2\operatorname{tr}(P^2)], \qquad \mathbb{E}T^2 \le \frac{499}{100}K^2L + \frac{1}{2}. \tag{S1.10}$$

Indeed $3\mathbb{E}T^2 = \mathbb{E}_g\mathbb{E}_X(g^TP^{1/2}X)^4$, and $\mathbb{E}_g\|P^{1/2}g\|^4 = 1 + 2\operatorname{tr}(P^2) \le 3$.

S1.4.2 A quadratic comparison for the exact lower shift

For $R \succ 0$, let $\Delta_- \ge 0$ be the unique number such that $R^{-1} + XX^T - \Delta_- I \succ 0$ and $\operatorname{tr}(R^{-1} + XX^T - \Delta_- I)^{-1} = \operatorname{tr} R$. Put

$$\eta = \|R\|_F, \quad P = R^2/\eta^2, \quad T = X^TPX, \quad \kappa = \|R\|/\eta.$$

We seek a positive matrix B for which

$$T \le (1 + X^TBX)\Delta_-. \tag{S1.11}$$

To prove this, we may suppose $X \ne 0$. Let $G = (R^{-1} + XX^T)^{-1}$ and $F = (G^{-1} - \Delta_- I)^{-1}$, and put

$$q = \operatorname{tr}(R - G) = \operatorname{tr}(F - G), \qquad c_X = \frac{X^TR^3X}{X^TR^2X}.$$

The rank-one inverse formula and Cauchy–Schwarz give

$$q = \frac{X^TR^2X}{1 + X^TRX} \le c_X \le \eta, \qquad \operatorname{tr}(G^2) = \eta^2 - 2qc_X + q^2.$$

Since $F - G = \Delta_- GF \succeq 0$,

$$\frac{q}{\Delta_-} = \operatorname{tr}(GF) \le \operatorname{tr}(G^2) + q\|G\|.$$

Using $q \le c_X$ and $\|G\| \le \|R\|$ gives $q/\Delta_- \le \eta^2 + q(\|R\| - c_X)$. Alternatively, $\|G\| \le \|G\|_F$ gives

$$\begin{aligned}\frac{q}{\Delta_-} &\le \eta^2 + q\left[q - 2c_X + \sqrt{\eta^2 - 2qc_X + q^2}\right] \\ &\le \eta^2 + q\left[\sqrt{\eta^2 - c_X^2} - c_X\right].\end{aligned}$$

The bracket in the first line increases with $q \in [0, c_X]$. Thus, with $z = c_X/\eta$,

$$\frac{q}{\Delta_-} \le \eta^2 + q\eta \left[\min\{\kappa, \sqrt{1-z^2}\} - z\right]. \tag{S1.12}$$

In particular, $B_0 = R + P(\kappa\eta I - R)$ satisfies (S1.11), by substituting $q = X^T R^2 X/(1 + X^T RX)$. More generally, $B = R + P(A\eta I - cR)$ does so whenever

$$\min\{\kappa, \sqrt{1-z^2}\} + (c-1)z \le A \qquad (0 \le z \le 1).$$

We will use this freedom when the spectrum is concentrated.

S1.4.3 From the mixed moment to a Laplace bound

The comparison reduces the probabilistic estimate to two quantities: the second moment of T and the mixed cost $\mathbb{E}(TX^T BX)$. The first determines the fluctuation constant; the second will determine the regularity condition and hence the correction constant.

Whenever such a matrix also satisfies $\mathbb{E}(TX^T BX) \le b$, let $m_2 \ge \mathbb{E}T^2$. Weighted Cauchy–Schwarz, as in [8, Lemma 4.2], and $e^{-x} \le (1+x)^{-1}$ give

$$\begin{aligned} \mathbb{E}e^{-\lambda(1+b)\Delta_-} &\le 1 - \lambda(1+b)\mathbb{E}\frac{T}{1 + X^T BX + \lambda(1+b)T} \\ &\le 1 - \frac{\lambda}{1+m_2\lambda} \le \exp\left(-\frac{\lambda}{1+(m_2-1/2)\lambda}\right) \qquad (\lambda > 0). \end{aligned} \tag{S1.13}$$

The last inequality follows by putting $z = \lambda/[2 + (2m_2 - 1)\lambda]$ in $\log[(1+z)/(1-z)] \ge 2z$. Taking $m_2 = (499/100)K^2 L + 1/2$, every mixed-cost cap proved below gives $\nu_- = (499/100)K^2 L$, and hence $\beta = 79/50$, since $499/100 < 2(79/50)^2$.

S1.5 Spectral costs and the lower correction budget

It remains to control the mixed cost without paying the worst spectral cost at every step of the barrier construction. We first obtain bounds depending on the spectral shape $\kappa = \|R\|/\|R\|_F$, then integrate that cost along the outward corrections. This is the step that yields $\alpha = 3.84$.

S1.5.1 A determinant envelope

Normalize $\eta = 1$. The eigenvalue of B_0 corresponding to an eigenvalue ρ of R is $b_0(\rho) = \rho + \rho^2(\kappa - \rho)$. For $0 < q < 1$ put

$$\mathcal{J}(q) = -\log(1-q) - q, \qquad \mathcal{M}(q) = q - \frac{2(1-q)}{q}\mathcal{J}(q) = \sum_{j=2}^{\infty} \frac{2q^j}{j(j+1)}.$$

If $q = u\kappa$ and $\mathcal{M}(q) \ge \kappa^2$, then

$$-\log[1 - ub_0(\rho)] - u\rho \le \frac{\rho^2}{\kappa^2}\mathcal{J}(u\kappa), \qquad 0 \le \rho \le \kappa. \tag{S1.14}$$

Here $\kappa - b_0(\rho) = (\kappa - \rho)(1 - \rho^2) \ge 0$, so the logarithm is well-defined. To prove the inequality in the strict case, set $c_0 = \mathcal{J}(q)/\kappa^2$, $d(\rho) = 1 - ub_0(\rho)$, and $\mathcal{F}(\rho) = u\rho + c_0\rho^2 + \log d(\rho)$. Then $\mathcal{F}(0) = \mathcal{F}(\kappa) = 0$ and

$$\begin{aligned} d(\rho)\mathcal{F}'(\rho)/\rho = (2c_0 - u^2 - 2u\kappa) + (3u - u^2\kappa - 2uc_0)\rho \\ + (u^2 - 2uc_0\kappa)\rho^2 + 2uc_0\rho^3. \end{aligned}$$

This cubic is positive at zero and negative at κ. Indeed the latter sign is precisely $\mathcal{M}(q) > \kappa^2$, and the former follows from $\mathcal{M}(q) < \mathcal{J}(q)/q - q/2$ by comparing their positive series. Its positive leading coefficient forces one root in $(0, \kappa)$ and another in (κ, ∞); the third root is negative, since their product is negative. Thus $\mathcal{F}$ first increases and then decreases on $[0, \kappa]$, proving $\mathcal{F} \ge 0$. The non-strict case follows by continuity.

S1.5.2 A uniform bound for concentrated spectra

When κ is large we use a comparison matrix adapted to that range, giving a constant upper bound for the mixed cost.

Lemma S1.5 (A uniform cost for concentrated spectra)**.** *Let $\|R\|_F = 1$, $\kappa = \|R\| \geq 71/100$, $P = R^2$ and $T = X^T P X$. There is a positive matrix B satisfying the lower-shift comparison $T \leq (1 + X^T B X)\Delta_-$ and*

$$\mathbb{E}(TX^T BX) - K^2 \operatorname{tr} R \leq \frac{4137}{1000} K^2 L.$$

Proof. First suppose $71/100 \leq \kappa \leq 147/200$, and set

$$s = \frac{347}{500}, \quad u = \frac{1207}{1000}, \quad w = \frac{26}{25}, \quad c = \frac{17}{10}, \qquad A(k) = k + \frac{7}{10}\frac{1 + s^2 - k^2}{2s}.$$

Take $B = R + P(A(\kappa)I - cR)$. The tangent inequality $\sqrt{1-k^2} \leq (1+s^2-k^2)/(2s)$ and (S1.12) give the required shift comparison, since

$$\min\{\kappa, \sqrt{1-z^2}\} + \frac{7}{10} z \leq \kappa + \frac{7}{10}\sqrt{1-\kappa^2} \leq A(\kappa).$$

For the first inequality split at $z = \sqrt{1-\kappa^2}$. On the left use $\kappa + (7/10)z$; on the right the function $\sqrt{1-z^2} + (7/10)z$ decreases, since $\kappa^2 \leq 100/149$.

We next give a scalar determinant majorant valid throughout this range. Set

$$A_0 = \frac{6}{5}, \quad A_1 = \frac{1211}{1000}, \quad v_0 = \frac{1631}{1000}, \quad v_1 = \frac{167}{100},$$
$$v(A) = v_0 + \frac{v_1 - v_0}{A_1 - A_0}(A - A_0).$$

The increasing polynomial $A(k)$ takes values in $[A_0, A_1]$ for $71/100 \leq k \leq 147/200$. For $A \in [A_0, A_1]$, define

$$d_A(\rho) = 1 - u(\rho + A\rho^2 - c\rho^3),$$
$$\psi_A(\rho) = u\rho + 2v(A)\rho^2 - 2w\rho^4,$$
$$g_A(\rho) = \psi_A(\rho) + \log d_A(\rho).$$

We claim that

$$d_A(\rho) > 0, \qquad g_A(\rho) \geq 0, \qquad 0 \leq \rho \leq 147/200. \tag{S1.15}$$

It suffices to check $A = A_0, A_1$: d_A is affine in A, and g_A is concave in A. Positivity of B follows at the same time from $1 + A\rho - c\rho^2 \geq 1 - c(147/200)^2 > 0$.

Here are exact certificates for the two endpoints. The derivative of $e^{\psi_A} d_A$ is $\rho e^{\psi_A} p_A(\rho)$, where the coefficients of p_A, in increasing order, are

$$4v - u^2 - 2uA, \quad 3uc - u^2 A - 4uv, \quad u^2 c - 4uvA - 8w,$$
$$4uvc + 8uw, \quad 8uwA, \quad -8uwc.$$

Write $b = 147/200$. For both endpoints, the coefficients of $(1+x)^5 p_A(bx/(1+x))$ have signs $+, +, +, -, -, +$. Moreover $p_A(0) > 0$, and its second root lies in $[r_-, r_+]$ with integer endpoint numerators, over the common denominator 10^5,

$$(63549, 63550) \quad \text{for } A_0, \qquad (64243, 64244) \quad \text{for } A_1.$$

Direct substitution gives $p_A(r_-) < 0 < p_A(r_+)$ and

$$d_A(r_+) T_{24}(\psi_A(r_-)) > 1.$$

The additional rational checks $1+2Ab-3cb^2>0$, $d_A(b)>0$ and $u+4vr_--8wr_+^3>0$ show that d_A decreases and is positive on $[0,b]$, while ψ_A increases on the root bracket. Descartes' rule therefore gives exactly two critical points, and the minimum of $e^{\psi_A}d_A$ is attained at zero or at the second one. This proves (S1.15).

Writing $\zeta=\operatorname{tr}(P^2)$, Gaussian comparison, with one largest eigenvalue kept exactly, now gives

$$\log\mathbb{E}\exp\left\{\frac{u}{2K^2}(X^TBX-K^2\operatorname{tr}R)\right\}\le v(A(\kappa))-w\zeta-\frac{1}{2}g_{A(\kappa)}(\kappa).$$

The fourth-moment bound gives $\mathbb{E}(T\log T)\le\log[5K^2L(1+2\zeta)/3]$. Maximizing $-w\zeta+\log[(1+2\zeta)/3]$ and then using $\sup_{L>0}(\log L+C)/L=e^{C-1}$ bounds the mixed cost by K^2L times

$$\frac{4}{u}+\frac{20}{3uw}\frac{e^{E(\kappa)}}{\sqrt{d(\kappa)}},\tag{S1.16}$$

where the explicit quartic polynomials are

$$\begin{aligned}E(k)&=v(A(k))(1-k^2)-4+\frac{w}{2}-\frac{uk}{2}+wk^4,\\ d(k)&=1-u(k+A(k)k^2-ck^3).\end{aligned}$$

The logarithmic derivative of the second term in (S1.16) has the sign of

$$Q(k)=2d(k)E'(k)-d'(k).$$

Every Bernstein coefficient of Q' on $[71/100,147/200]$ is less than -5; those of E' and d' are negative. Moreover $Q(k_-^*)>0>Q(k_+^*)$ for

$$k_-^*=\frac{72167}{10^5},\qquad k_+^*=\frac{72168}{10^5}.$$

Thus the maximum is attained between these two points. Since E and d decrease, its value is bounded by replacing $E(\kappa)$ by $E(k_-^*)$ and $d(\kappa)$ by $d(k_+^*)$. Finally, with $C=4137/1000$,

$$T_{24}(-2E(k_-^*))>\frac{400}{9w^2(uC-4)^2d(k_+^*)}.$$

This rational certificate proves that (S1.16) is less than C.

For $\kappa\ge147/200$, use $B_2=R+P((74/55)I-(19/10)R)$. It is positive, and it satisfies (S1.11), because $\sqrt{1-z^2}+(9/10)z\le\sqrt{181}/10<74/55$. Take

$$a=\frac{147}{200},\quad A=\frac{74}{55},\quad c=\frac{19}{10},\quad u=\frac{6}{5},\quad v=\frac{27}{16},\quad w=\frac{21}{20},$$

Define $d_2(\rho)=1-u(\rho+A\rho^2-c\rho^3)$, $\psi_2(\rho)=u\rho+2v\rho^2-2w\rho^4$, and $g_2=\psi_2+\log d_2$. The same derivative formula and Descartes test on $[0,1]$ have signs $+,+,-,-,+,+$, second-root bracket $[607/1000,608/1000]$, and

$$d_2(608/1000)T_{24}(\psi_2(607/1000))>1.$$

On that bracket d_2 decreases and ψ_2 increases. Thus $d_2>0$, $g_2\ge0$ on $[0,1]$, and g_2 increases above $608/1000$. Keeping its contribution at the largest eigenvalue therefore gives (S1.16) with $d=d_2(a)$ and $E=v-4+w/2-ua/2-va^2+wa^4$. The certificate

$$T_{24}(-2E)>\frac{400}{9w^2(uC-4)^2d}$$

proves the bound on this final range as well. □

S1.5.3 A spectral cost profile from entropy

For the remaining shapes, an entropy estimate gives a smaller cost that depends on κ. We combine it with the preceding uniform bound in the following profile.

Normalize $\|R\|_F = 1$, and retain $P = R^2$, $T = X^T P X$, $\kappa = \|R\|$ and $B_0 = R + P(\kappa I - R)$. Define

$$h_*(k) = \min\left\{\frac{4137}{1000}, \max\left\{\frac{3}{2}, \frac{57}{40} + \frac{23}{10}k + \frac{47}{20}k^2 - \frac{3}{10}k^3\right\}\right\}. \tag{S1.17}$$

We will prove the mixed-cost estimate

$$\mathbb{E}(TX^T B_0 X) - K^2 \operatorname{tr} R \le K^2 L h_*(\kappa) \tag{S1.18}$$

whenever $h_*(\kappa) < 4137/1000$. Together with Lemma S1.5, this gives the lower mixed-cost bound for every spectral shape.

The entropy estimate. For every real $p \ge 19/4$, put $\varepsilon_p = -\log(1 - e^{-(p-1/20)})$. Then

$$\sup_{y \ge 0} \frac{y^p}{e^y - 1} \le \frac{p^p e^{-p}}{1 - e^{-(p-1/20)}}. \tag{S1.19}$$

Indeed the unique positive maximizer y_p satisfies $y_p = p(1 - e^{-y_p})$. The inequality $e^{p-1/20} > 20p$, valid for $p \ge 19/4$, gives $y_p > p - 1/20$; now use $y^p e^{-y} \le p^p e^{-p}$. The initial exponential inequality follows from $T_{24}(47/10) > 95$ and the monotonicity of e^p/p.

The free moment order p lets us balance the entropy cost against the Gaussian determinant bound. Choose $0 < q < 1$, put $\mathsf{B} = 2\kappa^2/q$, and write $M = (1-q)^{-1/(2\kappa^2)}$. Gaussian comparison and $\operatorname{tr} P = 1$, $\|P\| = \kappa^2$ give $\mathbb{E}e^{T/(K^2\mathsf{B})} \le M$. Applying (S1.19) pointwise and then Jensen's inequality under the probability measure with density T gives

$$\begin{aligned} \mathbb{E}(T \log T) &\le \frac{\log \mathbb{E}T^p}{p-1} \\ &\le \frac{2p(L-1) + p\log(\mathsf{B}p) - p + \log(M-1) + \varepsilon_p}{p-1}. \end{aligned} \tag{S1.20}$$

Consequently, if $u\kappa < 1$, $\mathcal{M}(u\kappa) \ge \kappa^2$, and

$$(p-1)\frac{\mathcal{J}(u\kappa)}{2\kappa^2} + \log(M-1) + p\log(\mathsf{B}p) + \varepsilon_p \le 3p, \tag{S1.21}$$

then (S1.14), the entropy variational inequality, and (S1.20) imply

$$\mathbb{E}(TX^T B_0 X) - K^2 \operatorname{tr} R \le \frac{4p}{(p-1)u} K^2 L.$$

Thus the choice $u = 4p/[(p-1)h_*(\kappa)]$ proves (S1.18) once the displayed scalar conditions hold.

S1.5.4 Certifying the profile on the full spectral range

We now verify the scalar conditions for the chosen profile. The interval checks cover the middle range of κ; a direct bound covers small κ, and the uniform bound above covers the remaining concentrated spectra.

For $3/100 \le \kappa \le 711/1000$, use the cells $I_j = [j/1000, (j+1)/1000]$, $30 \le j < 711$. Writing $\bar{k}_j = (2j+1)/2000$ for the midpoint, choose the constant power

$$p_j = \begin{cases} \dfrac{37}{10} + \dfrac{13}{10\bar{k}_j} - \dfrac{\bar{k}_j^2}{2}, & 30 \le j < 650, \\ 12 - 10\bar{k}_j, & 650 \le j < 711. \end{cases}$$

Every power is at least 19/4. Put

$$D_p(k) = 1 + 2pk^2 - 4k^4, \quad q_p(k) = 1 - D_p(k)^{-1}, \quad \mathsf{B}_p(k) = 2k^2 + \frac{1}{p - 2k^2}.$$

These choices satisfy $\mathsf{B}_p = 2k^2/q_p$ and $\log M_p(k) = \log D_p(k)/(2k^2)$. The function B_p is increasing, and $\log M_p$ is decreasing: for the latter, $y \mapsto \log(1 + 2py - 4y^2)$ is concave and vanishes at zero. Also $\mathcal{J}(uk)/(2k^2)$ and $\mathcal{M}(uk)/k^2$ are increasing in both positive variables u, k, as follows from their positive power series.

For $x \geq 1$, let $U(x)$ be the logarithm upper bound in (S1.7) with $N = 24$ and $z = (x - 1)/(x + 1)$, rounded upwards to a multiple of 10^{-12}. On a cell $[a, b]$, let

$$u_+ = \frac{4p}{(p-1)h_*(a)}, \quad u_- = \frac{4p}{(p-1)h_*(b)}, \quad \ell = \frac{U(D_p(a))}{2a^2},$$

rounding ℓ upwards in the same manner. If $\ell \geq 8$, set $V(\ell) = \ell$. Otherwise let v be the reciprocal of the exponential upper bound in (S1.6), with $N = 40$ and $x = \ell$, rounded downwards to a multiple of 10^{-12}, and put

$$V(\ell) = \ell - \sum_{r=1}^{6} \frac{v^r}{r}.$$

Then $V(\ell) \geq \log(M_p(a) - 1)$. On every one of the 681 cells, exact rational arithmetic gives

$$u_+ b < 1, \qquad \sum_{r=2}^{60} \frac{2(u_- a)^r}{r(r+1)} \geq a^2,$$

$$(p-1)\frac{U((1-u_+b)^{-1}) - u_+b}{2b^2} + V(\ell) + pU(p\mathsf{B}_p(b)) + \frac{1}{T_{24}(p - 1/20) - 1} < 3p - \frac{1}{10000}.$$

The monotonicities above prove the required inequalities throughout each cell. These are finite rational checks, with all functions, subdivisions and powers specified explicitly.

For $0 < \kappa \leq 3/100$, take $p = 50$, $u = 400/147$, and use the same entropy and determinant bounds at the larger effective shape $k = 3/100$. Specifically, take $\mathsf{B} = \mathsf{B}_{50}(3/100)$; the bound $\|P\| \leq (3/100)^2$ gives the same Gaussian MGF bound, and $\mathcal{J}(u\kappa)/(2\kappa^2)$ is bounded by its value at k. Their certificate has the same strict margin $1/10000$. Here $u^2 > 3$, so $\mathcal{M}(u\kappa) \geq u^2\kappa^2/3 > \kappa^2$. This proves the remaining part of (S1.18). Finally the cubic in (S1.17) exceeds $4137/1000$ at $711/1000$, so Lemma S1.5 covers all remaining shapes.

S1.5.5 Integrating a spectral cost along the barrier

The next lemma converts a shape-dependent regularity condition into a total initial and outward cost. Its integral replaces a uniform bound for the profile, which would give a larger correction.

Lemma S1.6 (Cost of a mixed regularity condition)**.** *Let $a, b > 0$, and let $h : [0, 1] \to [0, \infty)$ be nondecreasing and absolutely continuous, with $h(k) > 0$ for $k > 0$ and $2\int_0^1 h(k)\,dk \geq h(1)$. A trace-preserving barrier construction imposing*

$$a \operatorname{tr} R + \|R\|_F h(\|R\|/\|R\|_F) \leq b$$

has total initial and outward cost at most

$$\frac{an + 2\mathcal{H}\sqrt{n}}{b}, \qquad \mathcal{H} = \int_0^1 \frac{kh(k)}{\sqrt{1-k^2}}\,dk,$$

when started at distance $[an+\sqrt{n}\,h(n^{-1/2})]/b$. *In particular, for* $h(k) = f + o\min\{k, k_0\}$, *where* $f, o \geq 0$, $0 \leq k_0 \leq 1$ *and* $f + ok_0 > 0$, *one has*

$$\mathcal{H} = f + o\mathcal{A}(k_0), \qquad \mathcal{A}(k_0) = \frac{\arcsin k_0 + k_0\sqrt{1-k_0^2}}{2}.$$

Proof. For $n > 1$, put $\tau = \operatorname{tr} R$, $F = \|R\|_F$, $M = \|R\|$. An outward shift has $R(\omega) = (R^{-1} + \omega I)^{-1}$. Both F and M/F decrease, since

$$\frac{d}{d\omega}\log\frac{M}{F} = -M + \frac{\operatorname{tr}(R^3)}{\operatorname{tr}(R^2)} \leq 0.$$

Thus the cap decreases continuously to zero and can always be restored. Cauchy–Schwarz on all eigenvalues except the largest gives

$$M \leq \frac{\tau}{n} + \sqrt{1-\frac{1}{n}}\sqrt{F^2 - \frac{\tau^2}{n}}.$$

For $n^{-1/2} \leq x \leq 1$, define

$$\psi_n(x) = \frac{1}{nx} + \sqrt{1-\frac{1}{n}}\sqrt{1-\frac{1}{nx^2}}, \qquad g(x) = a + xh(\psi_n(x)).$$

The function ψ_n is nondecreasing, and g is strictly increasing. Until a correction stops, $b < \tau g(F/\tau)$. Initially $\tau = b/g(n^{-1/2})$, and the cap already holds whenever $\tau \leq b/g(1)$. Since $-\tau' = F^2$ and the trace intervals of the corrections are disjoint, the sum of outward shifts is at most

$$\int_{b/g(1)}^{b/g(n^{-1/2})} \frac{d\tau}{\tau^2[g^{-1}(b/\tau)]^2} = \frac{1}{b}\int_{n^{-1/2}}^{1} \frac{g'(x)}{x^2}\,dx.$$

Adding the initial distance cancels the lower endpoint after integration by parts. With $\varphi = \arcsin(n^{-1/2})$ and $x = (\sqrt{n}\cos\theta)^{-1}$, the contribution beyond an/b, divided by $\sqrt{n}/b$, becomes

$$I_h(\varphi) = h(1)\sin\varphi + 2\int_0^{\pi/2-\varphi} h(\sin(\theta+\varphi))\sin\theta\,d\theta.$$

As $h' \geq 0$ almost everywhere,

$$\begin{aligned} I_h'(\varphi) &= -h(1)\cos\varphi + 2\int_0^{\pi/2-\varphi} h'(\sin(\theta+\varphi))\cos(\theta+\varphi)\sin\theta\,d\theta \\ &\leq \cos\varphi\left[-h(1) + 2\int_0^1 kh'(k)\,dk\right] = \cos\varphi\left[h(1) - 2\int_0^1 h(k)\,dk\right] \leq 0. \end{aligned}$$

The inequality uses $\sin\theta \leq \cos\varphi\sin(\theta+\varphi)$ and the substitution $k = \sin(\theta+\varphi)$. Hence $I_h(\varphi) \leq I_h(0) = 2\mathcal{H}$. For $n = 1$ there are no corrections, and the same formula follows directly. The stated special case satisfies the integral condition, and its integral is evaluated by splitting at k_0. □

S1.5.6 The general lower-tail constants

We apply the budget lemma to the certified profile and bound its integral by exact scalar comparisons.

The profile h_* is nondecreasing and absolutely continuous. Moreover $h_* \geq 3/2$ on $[0,1]$ and $h_* \geq 3$ on $[1/2,1]$, so $2\int_0^1 h_*(k)\,dk > h_*(1)$. Restoring the scale η gives a valid lower regularity cap

$$K^2\operatorname{tr} R + K^2 L\|R\|_F h_*(\|R\|/\|R\|_F) \leq b.$$

Lemma S1.6 bounds its correction coefficient by $\mathcal{H} = \int_0^1 k h_*(k)/\sqrt{1-k^2}\,dk$. For an exact bound, write $\Pi(k) = a_0 + a_1 k + a_2 k^2 + a_3 k^3$ for the cubic in (S1.17), and put $s = 713/1000$. On $[0, s]$, use $h_*(k) \le \Pi(k) + (3/40)\mathbf{1}_{\{k \le 1/30\}}$; on $[s, 1]$, use $h_* \le 4137/1000$. Integration gives

$$\mathcal{H} \le a_0 + \frac{2a_2}{3} + \frac{83}{80}\arcsin s + Q\sqrt{1-s^2} + \frac{3}{40}\left(1 - \sqrt{1-(1/30)^2}\right),$$

where

$$Q = \frac{4137}{1000} - a_0 - \frac{2a_2}{3} - \frac{a_2 s^2}{3} - \frac{a_1 s}{2} - \frac{a_3 s(3+2s^2)}{8} > 0.$$

The degree-seven alternating sine bound and two squared comparisons give

$$\arcsin s < \frac{793769}{10^6}, \qquad \sqrt{1-s^2} < \frac{701165}{10^6}, \qquad \sqrt{1-(1/30)^2} > \frac{999444}{10^6}.$$

Substitution yields

$$\mathcal{H} < \frac{96}{25}.$$

The transfer argument of Lemma S1.2, with dimension cost $K^2 n + 2K^2 L\mathcal{H}\sqrt{n}$, now proves the lower tail with

$$\alpha = \frac{96}{25} = 3.84, \qquad \beta = \frac{79}{50} = 1.58.$$

As before, the strict dimension estimate includes a zero target threshold.

S1.6 The lower-tail refinement when $K = 1$

We improve the mixed quadratic moment in the lower-shift comparison. Put $\eta = \|R\|_F$, $P = R^2/\eta^2$, $T = X^T P X$ and $\kappa = \|R\|/\eta$. The exact second moment and the fourth-order consequence of the directional MGF assumption suffice for the rounded constants.

S1.6.1 A mixed covariance bound

Compare the fourth-order coefficients in $\mathbb{E}\cosh(u\langle X, v\rangle) \le e^{u^2\|v\|^2/2}$, using its exact second moment, to obtain $\mathbb{E}\langle X, v\rangle^4 \le 3\|v\|^4$. This is the fourth-cumulant necessary condition for strict sub-Gaussianity; see [1, Proposition 3.2]. The Taylor argument applies here because the directional MGF assumption ensures all required moments. Gaussian averaging, as in (S1.10), therefore gives

$$\mathbb{E}T^2 \le 1 + 2\,\mathrm{tr}(P^2) \le 3, \qquad \mathrm{Var}(X^T A X) \le 2\|A\|_F^2 \quad (A \succeq 0).$$

For positive semidefinite A, B, this also gives

$$\mathrm{Cov}(X^T A X, X^T B X) \le \|A\|_F \|B\|_F + \mathrm{tr}(AB). \tag{S1.22}$$

Indeed, nonnegative variance implies $4\,\mathrm{Cov}(U, V) \le \mathrm{Var}(sU + s^{-1}V)$ for every $s > 0$. Apply the variance bound to $sA + s^{-1}B \succeq 0$ and minimize over s. Zero matrices follow by continuity. This reasoning does not assume independence of the coordinates.

S1.6.2 An orthogonal rank-one refinement

Suppose $P = auu^T + P_\perp$ and $B = duu^T + B_\perp$, where u is a unit vector, $a, d > 0$, and $P_\perp, B_\perp \succeq 0$ annihilate u. Put $p = \|P_\perp\|_F$, $q = \|B_\perp\|_F$. If $p, q > 0$ and

$$r = \frac{aq}{dp}, \qquad r + r^{-1} \le 6, \tag{S1.23}$$

then

$$\operatorname{Cov}(X^T P X, X^T B X) \le 2ad + 2pq. \tag{S1.24}$$

Here a, d denote the displayed eigenvalues only. To prove this, put $Z = \langle X, u\rangle^2$, $v = \operatorname{Var} Z$, $Y = X^T P_\perp X$, $W = X^T B_\perp X$, $x = \operatorname{Cov}(Z, Y)$ and $y = \operatorname{Cov}(Z, W)$, so $0 \le v \le 2$. Averaging the directional fourth-moment bound at $\sqrt{h}\,u + \sqrt{j}\,g$, with $g \sim N(0, P_\perp)$, gives

$$(2 - v)h^2 - 6xhj + 3(2p^2 - \operatorname{Var} Y)j^2 \ge 0 \qquad (h, j \ge 0).$$

Thus, writing $x_+ = \max\{x, 0\}$ and similarly for y_+,

$$\operatorname{Var} Y \le 2p^2 - \frac{3x_+^2}{2 - v}, \qquad \operatorname{Var} W \le 2q^2 - \frac{3y_+^2}{2 - v} \qquad (0 < v < 2).$$

If $x, y < 0$, ordinary Cauchy–Schwarz proves the claim immediately. Otherwise $xy \le x_+ y_+$. Positivity of the covariance matrix gives

$$\operatorname{Cov}(Y, W) \le \frac{xy}{v} + \sqrt{(\operatorname{Var} Y - x^2/v)(\operatorname{Var} W - y^2/v)}.$$

Using the two variance bounds and weighted arithmetic–geometric mean in the square root, with $\Lambda_v = (1 + v)/[v(2 - v)]$, bounds the full covariance by

$$adv + 2pq + dx_+ + ay_+ + \frac{x_+ y_+}{v} - \Lambda_v \left(\frac{q}{p} x_+^2 + \frac{p}{q} y_+^2\right).$$

Completing the square, the maximum of the last four terms over real x_+, y_+ is

$$ad(2 - v)\frac{(1 + v)(r + r^{-1}) + (2 - v)}{3(v + 4)} \le ad(2 - v).$$

This proves the claim for $0 < v < 2$. At $v = 0$, $x = y = 0$; at $v = 2$ the fourth-moment inequalities give $x, y \le 0$. Cauchy–Schwarz proves both endpoint cases directly.

S1.6.3 The cost when the spectrum is spread out

Suppose first that $\kappa \le 73/100$. Use the comparison $T \le (1 + X^T B_0 X)\Delta_-$ from Section S1.5, where $B_0 = R + P(\|R\| I - R)$. Normalize $\eta = \|R\|_F = 1$, write ρ_j for the eigenvalues of R, and put $\kappa = \max_j \rho_j$ and $s_\ell = \sum_i \rho_i^\ell$, so $s_2 = 1$. Bound (S1.22) gives

$$\mathbb{E}(T X^T B_0 X) - \operatorname{tr} R \le \kappa + \kappa s_4 - s_5 + \sqrt{s_4}\,\|B_0\|_F \le 2\kappa.$$

Here is the spectral inequality needed for the last step. The number $\kappa - \kappa s_4 + s_5$ is positive, and direct expansion gives

$$\begin{aligned}(\kappa - \kappa s_4 + s_5)^2 - s_4 \|B_0\|_F^2 \\ = (\kappa^2 - s_4)(1 - 2s_4) + \sum_{i<j} \rho_i^2 \rho_j^2 (\rho_i - \rho_j)^2 [2\kappa(\rho_i + \rho_j) - \rho_i^2 \rho_j^2].\end{aligned}$$

Every bracket is nonnegative. If $s_4 \le 1/2$, the whole expression is nonnegative, since $s_4 \le \kappa^2$. Otherwise put $v = \kappa^2 > 1/2$ and $r_* = \sqrt{1 - v}$. The largest eigenvalue is unique, all other eigenvalues r are at most r_*, and $2s_4 - 1 \le (2v - 1)^2$. The pair involving κ and r, divided by $r^2(v - r^2)$, equals

$$G(r) = v(\kappa - r)\left[2\kappa - \frac{v r^2}{\kappa + r}\right].$$

Here $\kappa - r \ge (2v - 1)/(2\kappa)$ and $v r^2/(\kappa + r) \le \kappa/2$, because $r^2 \le 1 - v < 1/2$. Hence $G(r) \ge (3v/4)(2v - 1) \ge (2v - 1)^2$, since $v \le (73/100)^2 < 4/5$. Finally, $\sum_{r \ne \kappa} r^2(v - r^2) = v - s_4$. Thus these pairs alone offset the negative first term. Taking square roots proves the cost 2κ in this range. The next argument covers $\kappa \ge 73/100$.

S1.6.4 A uniform cost for concentrated spectra

We next prove that, when $\kappa \geq 73/100$, a positive comparison matrix B can be chosen with

$$T \leq (1 + X^T BX)\Delta_-, \qquad \mathbb{E}(TX^T BX) \leq \operatorname{tr} R + \frac{73}{50}. \tag{S1.25}$$

For $c \geq 1$ and $A \geq \sqrt{1 + (c-1)^2}$, the tangent inequality $\sqrt{1-z^2} - z \leq A - cz$ gives the comparison $B = R + P(AI - cR)$. It is positive when $1 + A - c > 0$, by concavity of $1 + Ax - cx^2$ on $[0, 1]$. Convex combinations with B_0 remain valid comparison matrices. Bound (S1.22) and Young's inequality give, for $\xi > 0$,

$$\mathbb{E}(TX^T BX) - \operatorname{tr} R \leq A + \frac{1}{2\xi} + \sum_i \rho_i^2 g(\rho_i; A, c, \xi),$$

where

$$g(x; A, c, \xi) = (A/\xi + 1 - c)x + \left[A + \frac{\xi}{2} + \frac{A^2 - 2c}{2\xi}\right] x^2 - (c + Ac/\xi)x^3 + \frac{c^2}{2\xi}x^4.$$

First suppose $73/100 \leq k = \kappa \leq 7/9$ and let $p^2 = \sum_{\rho_i \neq k} \rho_i^4$ (the top eigenvalue is unique). If $p \leq 3/20$, the matrix B_0 alone suffices. Indeed, $s_3 \leq k^3 + \sqrt{1-k^2}\, p$, $s_5 \geq k^5$ and $s_6 \leq k^6 + (1-k^2)p^2$ give

$$\|B_0\|_F^2 \leq 1 + 2k\sqrt{1-k^2}\, p - p^2,$$

so its mixed cost is at most

$$k + kp^2 + \sqrt{(k^4 + p^2)(1 + 2k\sqrt{1-k^2}\, p - p^2)} < \frac{73}{50}.$$

For the last comparison replace k by $7/9$, p by $3/20$, and $2k\sqrt{1-k^2}$ by 1, then square the positive sides. The resulting rational inequality is verified by the checker.

If $p \geq 3/20$, take

$$B = \vartheta B_0 + (1-\vartheta)[R + P((73/55)I - (103/55)R)],$$
$$A = \vartheta k + (1-\vartheta)73/55, \qquad c = \vartheta + (1-\vartheta)103/55.$$

The tangent condition is exact: $(73/55)^2 = 1 + (48/55)^2$. Write $d = k(1 + Ak - ck^2)$ for the top eigenvalue of B, and q for the Frobenius norm of its remaining block. Here $A^2 \leq c$, $d \geq k^2$ and $q \leq (5/4)\sqrt{1-k^2} < 6/7$. The function $x^{-1} + A - cx$ decreases, so

$$1 \leq \frac{k^2 q}{dp} \leq \frac{40}{7}, \qquad \frac{k^2 q}{dp} + \frac{dp}{k^2 q} \leq \frac{1649}{280} < 6.$$

Thus (S1.24) applies. For any $\chi > 0$, its mixed-cost bound and $2pq \leq \chi p^2 + q^2/\chi$ give

$$\mathbb{E}(TX^T BX) - \operatorname{tr} R \leq A - ck^3 + 2k^2 d + \frac{1-k^2}{\chi} + \sum_{\rho_i \neq k} \rho_i^2 n(\rho_i; A, c, \chi),$$

where

$$n(x; A, c, \chi) = (2A/\chi - c)x + \left[\chi + \frac{A^2 - 2c}{\chi}\right] x^2 - \frac{2Ac}{\chi}x^3 + \frac{c^2}{\chi}x^4.$$

Define the two scalar majorants

$$O(k, x) = A + \frac{1}{2\xi} + k^2 g(k; A, c, \xi) + (1-k^2) g(x; A, c, \xi),$$
$$N(k, x) = A - ck^3 + 2k^2 d + \frac{1-k^2}{\chi} + (1-k^2) n(x; A, c, \chi).$$

Choose the parameters in the following table.

k interval	ϑ	ξ	χ	ω
$[73/100, 3/4]$	$9/20$	$8/5$	2	$3/10$
$[3/4, 7/9]$	$1/4$	$8/5$	$12/5$	$9/20$

For each row, the polynomial $73/50 - \omega O(k,x) - (1-\omega)N(k,x)$ has bidegree at most $(6,4)$. All its tensor-product Bernstein coefficients exceed $1/5000$ on each of the four rectangles obtained by taking $x \in [7j/40, 7(j+1)/40]$, $j = 0,1,2,3$. Thus 280 exact rational comparisons certify both rows. Every remaining eigenvalue is at most $\sqrt{1-k^2} < 7/10$; averaging with weights $\rho_i^2/(1-k^2)$ proves the desired mixed-cost bound.

For $k \geq 7/9$, take $A = 5/4$, $c = 7/4$ and $\xi = 8/5$. Then $A + 1/(2\xi) = 25/16$ and

$$g(x) = \frac{x}{32} + \frac{1849x^2}{1280} - \frac{399x^3}{128} + \frac{245x^4}{256}.$$

All Bernstein coefficients of $7/100 - g$ are positive on $[0, 1/4]$, $[1/4, 3/8]$, $[3/8, 1/2]$ and $[1/2, 1]$. If $J(k) = k^2 g(k) + (1-k^2)7/100$, the Bernstein coefficients of $-J'(k)/k$ on $[3/4, 1]$ are also positive, so J decreases there. Consequently the cost is at most

$$\frac{25}{16} + J(7/9) < \frac{73}{50}.$$

This completes the uniform-cost certificate.

S1.6.5 The lower correction budget

Restore the scale η and set

$$h(k) = \min\{2k, 73/50\}.$$

Use B_0 for $\kappa \leq 73/100$ and the uniform-cost comparison for $\kappa \geq 73/100$. They give the mixed-cost bound under the cap

$$\operatorname{tr} R + \eta h(\kappa) \leq b.$$

Lemma S1.6 applies with $a = 1$, $f = 0$, $o = 2$ and $k_0 = 73/100$. Put $k = 73/100$. Its singular-value correction is bounded by

$$2\mathcal{A}(k) \leq 2\left(k - \frac{k^3}{6} - \frac{k^5}{40}\right) < \frac{33}{25}.$$

Here we integrated $\sqrt{1-y^2} \leq 1 - y^2/2 - y^4/8$. The latter inequality follows by squaring the positive right side: $(1 - z/2 - z^2/8)^2 - (1-z) = z^3/8 + z^4/64 \geq 0$ for $0 \leq z \leq 1$. All the polynomial certificates and the final rational rounding are checked by the accompanying exact arithmetic script.

S1.6.6 The lower fluctuation bound

Under the same cap, $\mathbb{E}T^2 \leq 3$ allows us to take $m_2 = 3$ in (S1.13), giving

$$\mathbb{E}e^{-\lambda(1+b)\Delta_-} \leq \exp\left\{-\frac{\lambda}{1 + (5/2)\lambda}\right\}. \tag{S1.26}$$

Thus Lemma S1.2 gives $\beta = \sqrt{5/4} < 28/25$ under the same cap, while the correction budget gives $\alpha = 33/25$. These are the rounded $K = 1$ lower-tail constants in Theorem S1.1.

S1.7 The upper-tail constants

The exact upper shift is the soft upper spectral edge of $XX^T - R^{-1}$ at sensitivity $\operatorname{tr} R$, in the terminology of Srivastava and Vershynin [7, Section 1.9, equation (1.11)]. We prove the mean and exponential-moment bounds needed by Lemma S1.2 directly for the following parameters:

	w	ν_+
$K \geq 1$	$47/20$	$(9/2)K^2L$
$K = 1$	$58/25$	4

$L = \log(eK)$.

Here w is a dimensionless coefficient: the corresponding coefficient in the cap of Lemma S1.2 is wK^2L. Throughout this subsection $R \succ 0$, $\eta = \|R\|_F$, and $K^2 \operatorname{tr} R + wK^2L\eta \leq b < 1$. Let $\Delta_+ \geq 0$ be the unique shift satisfying

$$R^{-1} + \Delta_+ I - XX^T \succ 0, \qquad \operatorname{tr}(R^{-1} + \Delta_+ I - XX^T)^{-1} = \operatorname{tr} R.$$

Our assertions are $\mathbb{E}Y \leq 1$ for $Y = (1-b)\Delta_+$ and

$$\mathbb{E}e^{\lambda Y} \leq \exp\left(\frac{\lambda}{1-\nu_+\lambda}\right), \qquad 0 < \lambda < 1/\nu_+. \tag{S1.27}$$

The common spectral estimates come first, followed by the mean and exponential-moment arguments. The mean bound controls γ, while the MGF bound controls δ; both use the same regularity cap.

S1.7.1 Exact shifts and their tails

For $h \geq 0$, put

$$R_h = (R^{-1} + hI)^{-1}, \qquad B_h = hR_h + \frac{R_h^2}{\operatorname{tr}(RR_h)}.$$

The Sherman–Morrison calculation in (5.5) gives

$$\{\Delta_+ > h\} = \{X^T B_h X > h\}, \qquad X^T B_{\Delta_+} X = \Delta_+. \tag{S1.28}$$

Write r_j for the eigenvalues of R, and set

$$p_j(h) = \frac{r_j^2/(1+hr_j)}{\operatorname{tr}(RR_h)}.$$

These numbers sum to one, and the corresponding eigenvalue of B_h is $(hr_j + p_j(h))/(1+hr_j)$. For $0 < q < 1$,

$$1 - q\frac{hr_j + p_j(h)}{1+hr_j} = \frac{1+(1-q)hr_j}{1+hr_j}\left(1 - \frac{qp_j(h)}{1+(1-q)hr_j}\right).$$

The inequalities $\log(1+x) - \log(1+(1-q)x) \leq qx$ for $x \geq 0$ and $\prod_j (1-a_j) \geq 1 - \sum_j a_j$ for $a_j \geq 0$ with $\sum_j a_j \leq 1$ therefore give

$$\det(I - qB_h) \geq (1-q)e^{-qh\operatorname{tr} R}.$$

All the eigenvalue factors are positive. Apply the Gaussian comparison of Lemma S1.3 at $\lambda = q/(2K^2)$ and then Markov's inequality in (S1.28). Since $1 - K^2 \operatorname{tr} R \geq 1 - b > 0$, we obtain

$$\mathbb{P}\{Y > u\} \leq (1-q)^{-1/2} e^{-qu/(2K^2)}, \qquad u \geq 0, \quad 0 < q < 1. \tag{S1.29}$$

We also use a more precise tail when the eigenvalues of R are spread out. Put $\eta = \|R\|_F$ and $\kappa = \|R\|/\eta \in (0,1]$. The largest numerator $r_j^2/(1+hr_j)$ corresponds to $r_j = \|R\|$. Since

$\operatorname{tr}(RR_h) \ge \eta^2/(1+h\|R\|)$, we have $p_j(h) \le \kappa^2$ for every j. The concavity bound $\log(1-qv) \ge (v/\kappa^2)\log(1-q\kappa^2)$ for $0 \le v \le \kappa^2$ improves the determinant estimate to

$$\det(I - qB_h) \ge (1-q\kappa^2)^{1/\kappa^2} e^{-qh\operatorname{tr} R}.$$

This holds also at $q = 1$ when $\kappa < 1$. The same Gaussian comparison gives

$$\mathbb{P}\{Y > u\} \le (1-q\kappa^2)^{-1/(2\kappa^2)} e^{-qu/(2K^2)}. \tag{S1.30}$$

The margin in the cap allows a stronger version of this estimate:

$$\mathbb{P}\{Y > u\} \le (1-q_0\kappa^2)^{-1/(2\kappa^2)} e^{-q_0 u/(2K^2)} \tag{S1.31}$$

whenever $0 < q_0 \le 2$, $q_0\kappa^2 < 1$, and $u\kappa(q_0-1) \le wK^2L$. To prove it, put

$$\vartheta = \frac{1-K^2\operatorname{tr} R}{1-b} \ge 1 + \frac{wK^2L\eta}{1-b}, \qquad q = q_0/\vartheta, \qquad h = u/(1-b).$$

If $q \le 1$, the preceding determinant argument applies directly, with a prefactor no larger than the one in (S1.31). If $q > 1$, then $1 \le \vartheta < q_0 \le 2$ and

$$h\|R\| \le \frac{\vartheta - 1}{q_0 - 1} \le \frac{2-q}{q-1}, \qquad \frac{q}{1-(q-1)h\|R\|} \le q_0.$$

For the first comparison, after multiplying positive denominators the difference is $(q_0-1)(2-q_0) + (\vartheta-1)(\vartheta+q_0-2) \ge 0$. For the second, use $(q-1)h\|R\| \le (\vartheta-1)/\vartheta$. Now $\log(1+x) - \log(1-(q-1)x) \le qx$ for $0 \le x \le (2-q)/(q-1)$, as differentiation shows. Thus the first eigenvalue factors in the determinant still contribute at least $e^{-qh\operatorname{tr} R}$. The second factors have the form $1 - qp_j(h)/(1-(q-1)hr_j)$ and are at least $1 - q_0 p_j(h)$. Their product is therefore at least $(1-q_0\kappa^2)^{1/\kappa^2}$. Gaussian comparison and Markov's inequality give exponent $-q(1-K^2\operatorname{tr} R)h/(2K^2) = -q_0 u/(2K^2)$, which proves (S1.31).

Put $\rho_j = r_j/\eta$ and $a = \sum_j \rho_j^3$, so that $\kappa^3 \le a \le \kappa$. For $q > 0$ and $qk^2 < 1$, define

$$\mathcal{G}(k,a,q) = \exp\left\{\frac{q}{2} + \frac{a}{2k^3}[-\log(1-qk^2) - qk^2]\right\}. \tag{S1.32}$$

For $0 < q \le 2$, the tail comparison in (S1.31) improves to

$$\mathbb{P}\{Y > u\} \le \mathcal{G}(\kappa, a, q) e^{-qu/(2K^2)}, \qquad u\kappa(q-1) \le wK^2L. \tag{S1.33}$$

The sharper prefactor follows from majorization. Order the ρ_j decreasingly. The weights

$$p_j(h) = \frac{\rho_j^2/(1+h\eta\rho_j)}{\sum_\ell \rho_\ell^2/(1+h\eta\rho_\ell)}$$

are in the same order. Since the multiplier $(1+h\eta\rho_j)^{-1}$ increases with j, its weighted average on any initial segment is at most its weighted average on the complementary segment. Consequently every initial partial sum of $(p_j(h))$ is at most the corresponding sum of (ρ_j^2): the former vector is majorized by the latter. Convexity of $v \mapsto v^r$, for integers $r \ge 2$, gives

$$\sum_j p_j(h)^r \le \sum_j \rho_j^{2r} \le \kappa^{2r-4} \sum_j \rho_j^4 \le a\kappa^{2r-3}.$$

Expand $-\log(1-qp_j(h))$ into its positive power series and use $\sum_j p_j(h) = 1$. The resulting bound on the inverse square root of $\prod_j (1-qp_j(h))$ is exactly (S1.32). The proof of (S1.31) only

uses this product after bounding its second eigenvalue factors. Replacing that product bound proves (S1.33); its margin calculation is unchanged for every $K \ge 1$.

To extend the tail estimate beyond rate two, define the scalar excess

$$F_v(\theta) = \log \frac{\theta(\theta+v-2)}{(\theta-1)^2+v-1} - \frac{v(\theta-1)}{\theta(v-1)}.$$

The next lemma shows how an upper bound for this one-variable function controls the loss in the determinant estimate. Its finite certificates follow the proof.

Lemma S1.7 (Upper-tail rates exceeding two)**.** *Suppose $K^2 \operatorname{tr} R + wK^2L\eta \le b < 1$ and $0 < \kappa = \|R\|/\eta \le k$. Let $v > 2$, $vk^2 < 1$, $C_v \ge 0$, and suppose the scalar bound $F_v(\theta) \le C_v$ holds for $1 \le \theta \le v$. Then, for every $1 < v' \le v$,*

$$\mathbb{P}\{(1-b)\Delta_+ > u\} \le (1-vk^2)^{-1/(2k^2)} \exp\!\left(\frac{C_v}{2k^2} - \frac{v'u}{2K^2}\right) \quad \text{if } uk(v'-1) \le wK^2L. \tag{S1.34}$$

Proof. Set $Y = (1-b)\Delta_+$ and use the matrices B_h and weights $p_j(h)$ from (S1.28). Put

$$\vartheta = \frac{1-K^2 \operatorname{tr} R}{1-b}, \qquad h = \frac{u}{1-b}, \qquad q = \frac{v'}{\vartheta}.$$

The cap implies $\vartheta - 1 \ge wK^2L\eta/(1-b)$. If $q \le 1$, the existing determinant argument applies immediately, with a smaller prefactor than the one claimed. Otherwise $1 < \vartheta < v'$, and, with $x_0 = (\vartheta-1)/(v'-1)$, the assumed margin gives

$$h\eta \le x_0/k, \qquad hr_j \le (\rho_j/k)x_0 \le x_0, \qquad \rho_j = r_j/\eta.$$

All these numbers lie below $1/(q-1)$, because

$$(q-1)x_0 = \frac{(v'-\vartheta)(\vartheta-1)}{\vartheta(v'-1)} \le \frac{\vartheta-1}{\vartheta} < 1.$$

In particular, $q/[1-(q-1)hr_j] \le v'$. The second factors in the determinant factorization are therefore at least $1 - v'p_j(h) > 0$.

For the first factors define

$$F(q,x) = \log(1+x) - \log(1-(q-1)x) - qx.$$

One has $F(q,0) = F_x(q,0) = 0$ and

$$F_{xxx}(q,x) = \frac{2}{(1+x)^3} + \frac{2(q-1)^3}{(1-(q-1)x)^3} > 0.$$

The identity $F(q,x)/x^2 = \int_0^1 (1-\omega)F_{xx}(q,\omega x)\,d\omega$ shows that $F(q,x)/x^2$ increases with x. If $F(q,x_0) \ge 0$, it follows that

$$\sum_j F(q,hr_j) \le \sum_j \frac{\rho_j^2}{k^2} F(q,x_0) = \frac{F(q,x_0)}{k^2}.$$

If $F(q,x_0) < 0$, every summand is nonpositive, so the sum is at most zero. Thus in both cases the sum is at most $\max\{0, F(q,x_0)\}/k^2$.

A direct substitution gives

$$F(q,x_0) = F_{v'}(\vartheta).$$

To compare different rates, write $r = v - 1$ and $\upsilon = (\theta - 1)/(v - 1) \in [0, 1]$. The definition of F_v becomes

$$\log(1 + \upsilon) + \log(1 + r\upsilon) - \log(1 + r\upsilon^2) - \frac{(r + 1)\upsilon}{1 + r\upsilon},$$

whose derivative with respect to r equals $r\upsilon^2(1 - \upsilon)^2/[(1 + r\upsilon)^2(1 + r\upsilon^2)] \geq 0$. Therefore $F_{v'}(\vartheta) \leq C_v$ whenever $1 < v' \leq v$.

Using $\sum_j p_j(h) = 1$ and $p_j(h) \leq \kappa^2$, the second determinant factors have product at least $(1 - v'\kappa^2)^{1/\kappa^2}$. The logarithm of their inverse square root increases with both v' and κ, as follows by its positive power series, so the prefactor is at most $(1 - vk^2)^{-1/(2k^2)}$. The first factors contribute at most $qh \operatorname{tr} R/2 + C_v/(2k^2)$ to the logarithm of the Gaussian MGF bound. Markov's inequality now gives the claimed exponent, since

$$-\frac{q(1 - K^2 \operatorname{tr} R)h}{2K^2} = -\frac{v'u}{2K^2}.$$

For the two-piece entropy tail, take

$$s = \frac{w}{k(v - 1)}, \qquad A = (1 - vk^2)^{-1/(2k^2)} \exp\{C_v/(2k^2)\}.$$

The choice $v' = v$ gives $Ae^{-vu/(2K^2)}$ up to $u = sK^2L$. Above it choose $v' = 1 + s(v - 1)K^2L/u \in (1, v]$. The condition in (S1.34) holds with equality, and the bound becomes $Ae^{-u/(2K^2) - s(v-1)L/2}$. These two exponential pieces will be integrated in the entropy argument below. □

Certificates for the extended rates. The following choices satisfy $F_v(\theta) \leq C_v$ on $[1, v]$:

$$(v, C_v) = (81/40, 61/50000000), \quad (83/40, 57/2000000), \quad (17/8, 29/250000),$$

v	41/20	21/10	43/20	11/5	23/10	12/5
C_v	1/100000	13/200000	19/100000	1/2500	27/25000	21/10000
v	9/4	5/2	11/4	3	7/2	4
C_v	7/10000	7/2000	79/10000	27/2000	131/5000	201/5000

For a finite rational certificate, set $y = \theta - 1$, $V = vy$, $W = 2y^2 + vy + 2v - 2$ and $A_1 = (v - 1)(1 + y)$. Then $0 \leq V/W < 1$ and $F_v(1 + y) = \log[(1 + V/W)/(1 - V/W)] - vy/A_1$. Use

$$U(z) = S_6(z) + \frac{2z^{13}}{13(1 - z^2)} \geq \log\frac{1 + z}{1 - z},$$

by (S1.7). After multiplication by the positive denominator $A_1W^{11}(W^2 - V^2)$, the lower bound $C_v + vy/A_1 - U(V/W)$ has numerator

$$\begin{aligned} P_v(y) = {} & (C_vA_1 + vy)W^{11}(W^2 - V^2) \\ & - 2A_1\left[\sum_{j=0}^{5} \frac{V^{2j+1}W^{10-2j}(W^2 - V^2)}{2j + 1} + \frac{V^{13}}{13}\right]. \end{aligned}$$

This is a degree-27 polynomial. Starting with $[0, v - 1]$, test its Bernstein coefficients and bisect every interval whose coefficients are not all positive. At most eight bisections on any branch suffice for the fifteen displayed rates; every terminal interval has positive coefficients. The accompanying checker implements exactly this finite rational certificate.

S1.7.2 The comparison matrix and determinant bounds

The tail estimates alone do not give a sufficiently small mean cost. We next bound the shift by a fixed quadratic comparison. Its determinant excess J measures the cost that will be combined with entropy to prove $\mathbb{E}Y \le 1$.

Lemma S1.8 (Fixed upper comparison)**.** *Put $P = R^2/\eta^2$, $\bar r = \operatorname{tr}(RP)$, and $C = R + P(\bar r I - 2R)$. Then $C \succeq 0$, $\|C\| < \eta/2$, and*

$$B_h \preceq P + hC \quad (h \ge 0), \qquad \Delta_+ \le X^T P X + \Delta_+ X^T C X. \tag{S1.35}$$

For $a = \bar r/\eta$ the determinant excess $J = -\log\det(I - 2C/\eta) - 2\operatorname{tr} R/\eta$ satisfies

$$J \le \mathcal{J}_0(a) := -2a + \frac{671}{100}[1 - a^2 - a^2(1-a)^2].$$

Proof. The earlier fixed comparison uses $R_h^2 \preceq RR_h$, as in [10, Section 7.4, equation (7.3)]. Here we improve it using concavity of the individual eigenvalues of B_h. Write $b_j(h)$ for the eigenvalue corresponding to r_j. If $n = 1$, then $B_h = P = 1$ and $C = 0$, so the assertions are immediate. Otherwise put

$$S_j(h) = \sum_{k \ne j} \frac{r_k^2}{1 + h r_k}.$$

The eigenvalue formula gives

$$1 - b_j(h) = \frac{1}{1 + h r_j + r_j^2/S_j(h)}.$$

Cauchy–Schwarz yields $2S_j'(h)^2 \le S_j(h) S_j''(h)$, so $1/S_j$ is concave. The reciprocal of a positive concave function is convex; hence b_j is concave. Its tangent at zero gives

$$b_j(h) \le b_j(0) + h b_j'(0) = p_j + h(r_j + p_j(\bar r - 2r_j)), \qquad p_j = \frac{r_j^2}{\operatorname{tr}(R^2)}.$$

This proves the matrix comparison, and substituting $h = \Delta_+$ gives the shift bound. Since $\bar r \ge p_j r_j$, the eigenvalues of C are at least $r_j(1-p_j)^2 \ge 0$. Also $\operatorname{tr} C = \operatorname{tr} R - \bar r$.

Normalize $\eta = 1$, write $x_j = r_j/\eta$, and put $a = \sum_j x_j^3 = \bar r/\eta \in (0, 1]$. The normalized eigenvalues of C are $c_j = x_j + a x_j^2 - 2x_j^3$. We first record two norm bounds:

$$\sum_j c_j^2 \le 1 - a^2 - a^2(1-a)^2, \qquad \max_j c_j < \frac{223}{500}. \tag{S1.36}$$

For the first, put $p_j = x_j^2$ and define the symmetric nonnegative matrix

$$A_{jj} = (1 - p_j)^2, \qquad A_{jk} = p_j p_k \quad (j \ne k).$$

Then $c = Ax$ and the row and column sums of A are $1 - p_j$. Jensen's inequality after normalizing each nonzero row gives

$$\sum_j c_j^2 \le \sum_j (1 - p_j) \sum_k A_{jk} x_k^2 = 1 - 2s_2 - s_2^2 + 2s_3, \quad s_2 = \sum_j p_j^2, \quad s_3 = \sum_j p_j^3.$$

Now $s_3 \le s_2^{3/2}$ and $s_2 \ge a^2$. The function $u \mapsto 1 - 2u - u^2 + 2u^{3/2}$ decreases on $[0, 1]$, which proves the first bound in (S1.36). A row of sum zero contributes zero and requires no division.

For the second bound, fix $x = x_j$ and let $y = \sqrt{1 - x^2}$. Since $a \le x^3 + y^3$,

$$c_j \le x + x^2(x^3 + y^3) - 2x^3 = x y^3 (x + y).$$

Writing $u = x/y$ when $y > 0$, this is $u(1+u)/(1+u^2)^{5/2}$. Its derivative has the sign of $-(3u+1)(u^2+u-1)$, so its maximum is $((5+\sqrt{5})/10)^{5/2}$. The square of this maximum is

$$\frac{25+11\sqrt{5}}{250} < \frac{1241}{6250} < \left(\frac{223}{500}\right)^2.$$

Here $\sqrt{5} < 56/25$ suffices. The case $y = 0$ gives $C = 0$. This proves (S1.36), and in particular $I - 2C \succ 0$.

An analytic bound when a is large. Put $M = 223/500$. Expanding the logarithm and using $\sum c_j = \sum x_j - a$ and $c_j \le M$ gives

$$J \le -2a + \frac{-\log(1-2M) - 2M}{M^2}[1 - a^2 - a^2(1-a)^2].$$

The displayed coefficient is less than $671/100$. Indeed, $1 - 2M = 27/250$ and the positive Taylor sum for $e^{222672636/10^8}$ through degree eight exceeds $250/27$, while $222672636/10^8 = 2M + (671/100)M^2$. □

Six scalar majorants for the determinant excess. The preceding norm bound is useful for large a. On other spectral ranges we obtain smaller bounds by summing scalar logarithmic majorants. Each is used only on the domain certified below.

Write $x_j = r_j/\eta$ and $a = \sum_j x_j^3$. All the following scalar formulas use the normalization $\sum_j x_j^2 = 1$. Set $v_a = a - 7/20$ and

i	$\mathcal{B}_i(a)$	$\mathcal{A}_i(a)$
1	$13/6$	$1 + 9a$
2	$13/6 - 4(a - 1/4)^2$	$20a - 2$
3	$13/5$	$24a - 22/3$
4	$13/6 - 4(a - 1/4)^2 - 1/30$	$20a - 2$
5	$13/6 - 1/40$	$1 + 9a$

$$\mathcal{B}_6(a) = \frac{413}{200} - \frac{1339}{1000}v_a - \frac{10009}{1000}v_a^2 - \frac{25197}{1000}v_a^3,$$
$$\mathcal{A}_6(a) = \frac{816}{125} + \frac{7307}{200}v_a + \frac{9971}{100}v_a^2 + \frac{18903}{250}v_a^3,$$
$$\mathcal{H}_i(a,x) = 2x + \mathcal{B}_i(a)x^2 - \mathcal{A}_i(a)x^2(x-a), \qquad z(a,x) = 2(x + ax^2 - 2x^3).$$

The six inequalities

$$-\log(1 - z(a,x)) \le \mathcal{H}_i(a,x) \tag{S1.37}$$

hold on the domains in the table below. Their common rational certificate uses

$$\mathcal{D}(z) = 1 - \frac{16}{7}z + \frac{12}{7}z^2 - \frac{16}{35}z^3 + \frac{1}{35}z^4,$$
$$\mathcal{N}(z) = z - \frac{25}{14}z^2 + \frac{19}{21}z^3 - \frac{47}{420}z^4.$$

Indeed,

$$\mathcal{D}(z) = \frac{1-z}{35}[4 + 18(1-z) + 12(1-z)^2 + (1-z)^3] > 0 \qquad (z < 1),$$

and direct differentiation gives

$$\frac{d}{dz}\left[\frac{\mathcal{N}(z)}{\mathcal{D}(z)} + \log(1-z)\right] = \frac{z^7}{1225\mathcal{D}(z)^2}.$$

The bracket vanishes at zero and is nonnegative on either side of zero. Consequently $-\log(1-z) \le \mathcal{N}(z)/\mathcal{D}(z)$ for $z < 1$. On all domains below, $z < 1$: for $x \le 1/2$ the largest cubic $x + (5/6)x^2 - 2x^3$ is at most $11/24$, and for $x = 1/2 + u$ it is at most $11/24 + u/3 - 2u^2 \le 17/36$.

The polynomials $\mathcal{Q}_i = (\mathcal{H}_i\mathcal{D}(z) - \mathcal{N}(z))/x^2$ extend to $x = 0$ by cancellation. The following table specifies their initial rectangles, bidegrees, and the number of terminal rectangles in a single uniform certificate:

i	a interval	x interval	bidegree	rectangles
1	$[0, 27/40]$	$[0, 1]$	$(6, 13)$	19
2	$[1/4, 37/50]$	$[0, 1]$	$(6, 13)$	19
3	$[7/10, 5/6]$	$[0, 1]$	$(6, 13)$	13
4	$[1/4, 7/10]$	$[0, 1]$	$(6, 13)$	28
5	$[0, 13/20]$	$[0, 1]$	$(6, 13)$	16
6	$[1/4, 51/100]$	$[0, 13/20]$	$(8, 13)$	70

Use the tensor-product Bernstein coefficients defined above. Starting from each displayed domain, retain a rectangle if all its coefficients are positive; otherwise bisect both coordinates and test all four children. The exact checker reproduces this deterministic rule. It stops within five bisections on every branch, with the displayed counts, and every terminal coefficient exceeds $1/100000$. This proves all six scalar majorants.

Summing $\mathcal{H}_i(a, x_j)$ cancels the term containing $x_j - a$. Retaining the scalar excess at a largest eigenvalue gives

$$J \le \mathcal{B}_i(a) - \mathcal{H}_i(a, \kappa) + \frac{\mathcal{N}(z(a, \kappa))}{\mathcal{D}(z(a, \kappa))}. \tag{S1.38}$$

Use a majorant only on its stated a interval; the sixth one additionally requires $\kappa \le 13/20$. For $a \ge 5/6$ the norm bound $\mathcal{J}_0(a)$ supplies the alternative certificate. In particular, summing the fifth majorant gives $J \le 257/120$ for $a \le 13/20$.

We first record an additional determinant estimate for concentrated spectra. If $\kappa \ge 37/40$, every eigenvalue ρ_i except a largest one is less than $381/1000$, since $1 - \kappa^2 \le 231/1600 < (381/1000)^2$. As $a \le 1$, its contribution to C/η is at most $f(\rho_i)$, where $f(x) = x + x^2 - 2x^3$. This function increases on $[0, 381/1000]$ and $f(381/1000) < 21/50$. The largest contribution is at most $\kappa + \kappa^2 - 2\kappa^3 \le 3(1 - \kappa) \le 9/40$. Thus $\|C\|/\eta \le M := 21/50$. The Frobenius-norm argument in Lemma S1.8 yields

$$J \le \mathcal{J}_1(a) := -2a + \frac{563}{100}[1 - a^2 - a^2(1 - a)^2] \qquad (\kappa \ge 37/40). \tag{S1.39}$$

Indeed, the positive degree-24 Taylor sum proves

$$e^{2M + (563/100)M^2} > 25/4 = 1/(1 - 2M),$$

which gives the claimed logarithmic coefficient.

S1.7.3 Entropy bounds and the upper mean

We now combine the determinant estimates with the shift tails. The common reduction below shows that it suffices to bound an entropy cost plus $J/2$; the two mean lemmas then certify this bound at $K = 1$ and for general K.

For any mean-one $W \ge 0$, with $0 \log 0 = 0$, write $D(W) = \mathbb{E}W \log W$. For $c > 0$, the part below c of $W \log(W/c)$ is nonpositive, so

$$D(W) \le \log c + \int_c^\infty [1 + \log(u/c)]\mathbb{P}\{W > u\}\, du. \tag{S1.40}$$

For later integration put $I(x) = \int_0^\infty e^{-u}/(x + u)\, du$, $x > 0$. The entropy integral is bounded using

$$\frac{x^2 + 8x + 11}{x^3 + 9x^2 + 18x + 6} \le I(x) \le \frac{x^3 + 11x^2 + 26x + 6}{x(x^3 + 12x^2 + 36x + 24)}.$$

For completeness, with U exponential of mean one, $D_0 = x^3 + 9x^2 + 18x + 6$ and $D_1 = x^3 + 12x^2 + 36x + 24$, the exact remainders are

$$I(x) - \frac{x^2 + 8x + 11}{D_0} = \frac{1}{D_0^2}\mathbb{E}\frac{(U^3 - 9U^2 + 18U - 6)^2}{x + U},$$
$$\frac{x^3 + 11x^2 + 26x + 6}{xD_1} - I(x) = \frac{1}{xD_1^2}\mathbb{E}\frac{U(U^3 - 12U^2 + 36U - 24)^2}{x + U}.$$

Expanding these identities uses only $\mathbb{E}U^j = j!$; the accompanying checker verifies the polynomial and factorial-moment identities exactly.

For both upper-mean certificates, use (S1.6)–(S1.7) with 18 logarithmic terms after reduction by powers of two, and exponential Taylor degree 80. Round intermediate bounds outwards to multiples of 10^{-15}. On the $K = 1$ integration grid, successive $e^{-qu_j/2}$ values use multiplication by an upper bound for $e^{-q/400}$ and upward rounding. The final rounding precisions are specified below.

The same final step converts both entropy estimates below into mean bounds. If $\mu = \mathbb{E}Y \geq 1$, then $W = Y/\mu \leq Y$ has mean one and inherits all its tail bounds. If $D(W) + J/2 \leq wL$, Gaussian comparison and (S1.5) give

$$\mathbb{E}(WX^TCX) \leq K^2\eta\big(D(W) - \tfrac{1}{2}\log\det(I - 2C/\eta)\big) \leq K^2 \operatorname{tr} R + wK^2L\eta \leq b.$$

The fixed comparison then gives $\mu \leq 1 - b + b\mu$, hence $\mu \leq 1$. The tail family ensures that μ is finite.

Lemma S1.9 (Upper mean when $K = 1$)**.** *For $K = 1$ the cap $\operatorname{tr} R + (58/25)\eta \leq b < 1$ implies $\mathbb{E}Y \leq 1$.*

Proof. Set $w = 58/25$. By the preceding reduction it suffices to prove

$$D(W) + J/2 \leq w \tag{S1.41}$$

for every mean-one weight satisfying (S1.33). First suppose $0 < \kappa \leq 1/4$. The prefactor in (S1.32) is at most $(1 - q\kappa^2)^{-1/(2\kappa^2)}$, because $a \leq \kappa$. On each of the fifty intervals $[i/200, (i + 1)/200]$, write $k_i = (i + 1)/200$ and choose c_i by rounding $43/20 + (23/10)k_i^2$ upwards to a multiple of $1/200$. Use the parameters $q \in \{j/20 : 18 \leq j \leq 40\}$ satisfying $qk_i^2 < 1$ and $u_jk_i(q - 1) \leq w$, with $u_j = j/200$. The upper Riemann sum for (S1.40) uses the minimum of these tail bounds and 1 at u_j, multiplied by $[1 + \log(u_{j+1}/c_i)]/200$. Integrate this sum to 40. For the remaining integral use $q = 9/10$, $\sqrt{10} < 31623/10000$ and $\log(u/c_i) \leq u/c_i$. Writing $\omega = 9/20$, the remainder is at most

$$\mathcal{R}_0(c) = \frac{31623}{10000}e^{-40\omega}\left(\frac{1}{\omega} + \frac{40}{c\omega} + \frac{1}{c\omega^2}\right). \tag{S1.42}$$

After adding $\log c_i$, each resulting bound lies more than $1/500$ below

$$\frac{239}{200} + \frac{9}{20}\left(\frac{i}{200}\right)^2 + \frac{1}{2}\left(\frac{i}{200}\right)^4.$$

The polynomial is increasing. Together with $J \leq 13/6 - 1/40$, proved above, this gives

$$D(W) + \tfrac{1}{2}J \leq \frac{239}{200} + \frac{9}{20}\left(\frac{1}{4}\right)^2 + \frac{1}{2}\left(\frac{1}{4}\right)^4 + \frac{13}{12} - \frac{1}{80} < \frac{58}{25}.$$

For $\kappa \geq 1/4$, use the following joint certificate. Partition $[1/4, 1]$ at $k = i/80$ for $20 \leq i \leq 80$. Partition $[0, 1]$ at all $j/40$, adding $37/50$ and $5/6$. Discard rectangles $[a_-, a_+] \times [k_-, k_+]$ for

which $a_+ < k_-^3$ or $a_- > k_+$. Increase every remaining lower a endpoint to $\max\{a_-, k_-^3\}$ and discard empty intervals. The resulting 837 rectangles still cover all admissible pairs. The joint polynomial test below succeeds on all 837 rectangles without any further subdivision.

For each rectangle, let $\bar{k}, \bar{a}$ denote its midpoints and put $v = \bar{k} - 1/2$. Set

$$c_0 = \begin{cases} 43/20 + (6/5)\bar{k}^2 + (9/2)\bar{k}^4, & \bar{k} \le 1/2, \\ 273/100 + (37/10)v + 17v^2 - 11v^3, & \bar{k} > 1/2. \end{cases}$$

Round $c_0 - (2\bar{k} + 1/4)(\bar{k} - \bar{a})$ upwards to the next multiple of $1/200$ to obtain c. For $u_j = j/200$, $200c \le j < 8000$, put

$$q_j^* = \frac{u_j - 1}{\bar{k}\bar{a} + \bar{k}^2(u_j - 1)},$$
$$q_j = \frac{1}{100} \max\left\{90, \min\left\{200, \lfloor 100 q_j^* \rfloor, 100 + \left\lfloor \frac{100w}{u_j k_+} \right\rfloor, \left\lceil \frac{100}{k_+^2} \right\rceil - 1 \right\}\right\}.$$

These parameters satisfy all restrictions in (S1.33) at u_j throughout the rectangle. Define

$$E(a, k) = \log c + \frac{1}{200} \sum_{j=200c}^{7999} [1 + \log(u_{j+1}/c)] \mathcal{G}(k, a, q_j) e^{-q_j u_j/2} + \mathcal{R}_0(c).$$

The monotonicity of survival probabilities gives $D(W) \le E(a, \kappa)$. The separate convexity of E is useful here: indeed,

$$\log \mathcal{G}(k, a, q) = \frac{q}{2} + \frac{a}{2} \sum_{r=2}^{\infty} \frac{q^r k^{2r-3}}{r}$$

is affine in a and convex in k, with nonnegative coefficients. Exponentiation preserves both separate convexities. Thus E lies below the bilinear interpolation of its four corner values. Bound those values upwards, rounding finally to multiples of 10^{-9}, and denote their bilinear interpolation by $\mathcal{E}_0$.

On each rectangle, use any one of the applicable bounds (S1.38). The checker verifies that all Bernstein coefficients of at least one applicable polynomial

$$\left[w - \mathcal{E}_0 - \frac{\mathcal{B}_i}{2} + \frac{\mathcal{H}_i}{2}\right] \mathcal{D} - \frac{\mathcal{N}}{2}$$

are strictly positive. On rectangles with $a_- \ge 5/6$, it can instead verify the polynomial $w - \mathcal{E}_0 - \mathcal{J}_0/2$ with strictly positive coefficients. Since the relevant denominator $\mathcal{D}$ is positive, these checks establish (S1.41). The rational logarithm and exponential rules above evaluate all corner bounds; both spectral ranges use the same $1/200$ integration mesh. The common mean reduction completes the proof. □

Lemma S1.10 (Upper mean for general K)**.** *Let $K \ge 1$, $L = \log(eK)$ and $w = 47/20$. In the notation of the upper-shift proof, suppose*

$$K^2 \operatorname{tr} R + wK^2 L\eta \le b < 1, \qquad \eta = \|R\|_F.$$

Then $Y = (1 - b)\Delta_+$ has mean at most one.

Proof. Set $w = 47/20$. On a spectral rectangle with upper endpoints k, a_+, use the tail prefactor

$$A = \mathcal{G}(k, a_+, q) e^{C_q/(2k^2)}, \qquad s = \frac{w}{k(q-1)} \quad (q > 1),$$

where $C_q = 0$ for $q \le 2$ and the rate certificate supplies it for $q > 2$. The positive series for $\log \mathcal{G}$ shows that it increases in each argument, so this prefactor works also for the adaptive rate above the cutoff. Integrating the resulting tails in (S1.40), with cutoff cK^2L, gives

$$D(W) \le 2L - 2 + \log(cL) + \mathcal{E}(L), \qquad \mathcal{E} = \mathcal{R} + \mathcal{T},$$

where

$$\mathcal{R}(L) = \frac{2A}{q} e^{-2-(qc/2-2)L}[1 + I(qcL/2)],$$
$$\mathcal{T}(L) = Ae^{-2-(qs/2-2)L}\left[(2 - 2/q)(1 + \log(s/c)) + 2I(sL/2) - \frac{2}{q} I(qsL/2)\right].$$

Set $\mathcal{T} = 0$ for $q \le 1$. Every witness has $qc/2 > 2$ and, for $q > 1$, also $s \ge c$ and $qs/2 > 2$. Thus $\mathcal{E}$ is decreasing and convex. For $\mathcal{R}$ this follows from the integral defining I; for $\mathcal{T}$ use

$$I(x) - \frac{1}{q} I(qx) = \int_0^\infty e^{-xv}\left[\frac{1}{1+v} - \frac{1}{q(q+v)}\right] dv \qquad (q > 1).$$

The kernel is nonnegative, and multiplication by a decreasing exponential preserves these properties.

For $\kappa \le 2/5$ we use the fifth scalar majorant $\mathcal{H}_5$, which strengthens the first and is already proved for $a \le 13/20$. Summing it cancels the term containing $\rho_i - a$, and gives $J \le 13/6 - 1/40 = 257/120$. Use $k = a_+ = 2/5$, $q = 11/4$ and $c = 193/100$ in this range. The entropy calculation below has margin greater than $1/100000$.

For $2/5 \le \kappa \le 1$, start with the intervals $[i/80, (i+1)/80]$, $32 \le i \le 79$. Partition $a \in [0, 1]$ at the multiples of $1/40$, adding $27/40$, $37/50$ and $5/6$. On each rectangle $[k_-, k_+] \times [a_-, a_+]$, impose the spectral restrictions by replacing k_- by $\max\{k_-, a_-\}$, then a_- by $\max\{a_-, k_-^3\}$ with this updated endpoint, and a_+ by $\min\{a_+, k_+\}$. Discard rectangles with empty interior; boundary cases follow by continuity. Some rectangles are subdivided at both midpoints, with the same restrictions imposed on each child. At most five such bisections on any branch suffice. The exact checker supplies and verifies rational witnesses on the resulting 1330 rectangles, and checks that these rectangles cover every initial rectangle after the restrictions.

A witness consists of a rational determinant bound J_*, an applicable scalar majorant, a rate q, and a cutoff coefficient c. The permitted rates are the multiples of $1/40$ in $[9/10, 2]$ together with the fifteen rates in the extension certificate. The cutoffs are positive multiples of $1/100$ not exceeding 8. For each of the six scalar majorants, the determinant certificate is positivity of every Bernstein coefficient of

$$[J_* - \mathcal{B}_i(a) + \mathcal{H}_i(a, \kappa)]\mathcal{D}(z(a, \kappa)) - \mathcal{N}(z(a, \kappa))$$

on the rectangle, with the majorant used only on its established domain. The coefficients of $\mathcal{D}$ are also checked positive after degree elevation to the same bidegree. Alternatively, when $a_- \ge 5/6$ use the polynomial $J_* - \mathcal{J}_0(a)$ from the earlier norm bound; when $k_- \ge 37/40$ one may use $J_* - \mathcal{J}_1(a)$ from (S1.39). These finite checks prove $J \le J_*$ throughout every rectangle. Use the upper bound for positive coefficients of I and the lower bound for its negative coefficient in $\mathcal{T}$. At $L = j/40$, $40 \le j \le 160$, apply the rational logarithm and exponential rules above, rounding intermediate bounds outwards to multiples of 10^{-15} and final upper bounds for $\mathcal{E}(L)$ to multiples of 10^{-10}. Write e_ℓ, e_h for consecutive such bounds and $\sigma = (e_h - e_\ell)/(h - \ell)$. Convexity bounds $\mathcal{E}$ by the chord joining them. For $r = w - 2 - \sigma > 0$ put $x_* = \min\{h, \max\{\ell, 1/r\}\}$. The finite comparison

$$2 + rx_* - \log(cx_*) - \frac{J_*}{2} - e_\ell + \sigma\ell > \frac{1}{100000}$$

then proves $D(W) + J/2 < wL$ on $[\ell, h]$. For $L \geq 4$, use $\mathcal{E}(L) \leq e_4$ and $1/(w-2) < 4$; the remaining comparison is

$$2 + 4(w-2) - \log(4c) - \frac{J_*}{2} - e_4 > \frac{1}{100000}.$$

The checker verifies these inequalities for every rational witness. This proves the required entropy bound for all $L \geq 1$ and all spectra.

The common mean reduction completes the proof. □

S1.7.4 Exponential moments

With the mean bounds established, we turn to ν_+ and the fluctuation constant. A first-moment bound and an exponential tail control all higher moments. The second moment needs a separate refinement, especially near $K = 1$ and for concentrated spectra.

Lemma S1.11 (Moments from a first moment and an exponential tail)**.** *Suppose $Y \geq 0$, $\mathbb{E}Y \leq 1$, and $\mathbb{P}\{Y > u\} \leq \sqrt{10}e^{-9u/(20K^2)}$. Then $\mathbb{E}Y^p/p! \leq [(9/2)K^2L]^{p-1}$ for every integer $p \geq 3$. For $L \geq 5/4$ one also has $\mathbb{E}Y^2/(2K^2L) - 1/(2K^2L) < 9/2$.*

Proof. We combine the first moment and the tail as in [5, Lemma 2.1]. Put $A_* = 15K^2L/2$, $s_0 = 20K^2/9$, and $\varepsilon_0 = \sqrt{10}\, s_0 e^{-A_*/s_0}$. Since $L \geq 1$,

$$\varepsilon_0 = \frac{20\sqrt{10}}{9} e^{-2-11L/8} \leq \frac{1}{4} e^{-11(L-1)/8} \leq \frac{1}{4}.$$

For the first inequality, use $\sqrt{10} < 3163/1000$ and $e^{27/8} > 146/5$; the latter follows from the positive Taylor sum through degree fourteen. For each integer $p \geq 2$, the function $u \mapsto u^p - A_*^{p-1}u$ is nonpositive below A_*. Integrating its positive part against the tail and using $\mathbb{E}Y \leq 1$ gives

$$\frac{\mathbb{E}Y^p}{p!} \leq (1-\varepsilon_0)\frac{A_*^{p-1}}{p!} + \varepsilon_0 \sum_{j=0}^{p-1} \frac{A_*^j s_0^{p-1-j}}{j!}. \tag{S1.43}$$

The coefficient of ε_0 is positive. Since $s_0 \leq (20/9)K^2L$, it follows that

$$\frac{\mathbb{E}Y^p}{p!} \leq (K^2L)^{p-1} u_p, \qquad u_p = \frac{3}{4}\frac{(15/2)^{p-1}}{p!} + \frac{1}{4}\sum_{j=0}^{p-1} \frac{(15/2)^j (20/9)^{p-1-j}}{j!}.$$

Direct rational arithmetic gives

$$u_3 = \frac{25225}{1296} < \left(\frac{9}{2}\right)^2, \qquad u_4 = \frac{10896875}{186624} < \left(\frac{9}{2}\right)^3.$$

For $p \geq 4$, the first component grows by the factor $(15/2)/(p+1)$. Writing T_p for the sum in the second component,

$$T_{p+1} = \frac{20}{9}T_p + \frac{(15/2)^p}{p!} \leq \left(\frac{20}{9} + \frac{15}{2p}\right) T_p \leq \frac{295}{72}T_p < \frac{9}{2}T_p.$$

This proves the higher-moment assertion.

For the second moment the same integration gives

$$\frac{\mathbb{E}Y^2}{2K^2L} - \frac{1}{2K^2L} \leq f_0(L) := \frac{15}{4} + \frac{1}{4}e^{-11(L-1)/8}\left(\frac{15}{4} + \frac{20}{9L}\right) - \frac{e^{-2(L-1)}}{2L}.$$

For $L \geq 1$,

$$-L^2 e^{2(L-1)} f_0'(L) \geq \frac{165}{128}L^2 - \frac{17}{72}L + \frac{1}{18} > 0.$$

The polynomial has positive coefficients in $L-1$. The bounds (S1.6) with $N = 24$ at $11/32$ and $1/2$ give $f_0(5/4) < 9/2$. These are rational checks in the accompanying script. □

Lemma S1.12 (Upper exponential moment when $K = 1$)**.** *Assume $K = 1$ and $\operatorname{tr} R + w\eta \le b < 1$, where $\eta = \|R\|_F$ and $w \ge 23/10$. Suppose the normalized upper shift $Y = (1-b)\Delta_+$ satisfies $\mathbb{E}Y \le 1$. Then* (S1.27) *holds with $\nu_+ = 4$.*

Proof. Put $\kappa = \|R\|/\eta$. It suffices to bound the moment generating function by $1 + \lambda + \sum_{p\ge2} u_p\lambda^p$, where $u_2 \le 9/2$ and $u_p \le 4^{p-1}$ for every integer $p \ge 3$. Indeed, this series is at most

$$1 + \frac{\lambda}{1-4\lambda} + \frac{\lambda^2}{2} \le \exp\left(\frac{\lambda}{1-4\lambda}\right) \qquad (0 < \lambda < 1/4).$$

First suppose $\kappa \le 15/16$ and put $v = 225/256$. The weights $p_j(h)$ in the determinant argument obey $0 \le p_j(h) \le v$ and $\sum_j p_j(h) = 1$. For any such weights, $\prod_j(1-p_j(h)) \ge v(1-v)$. Indeed, replacing two weights by $0, a+b$ when $a+b \le v$, or by $a+b-v, v$ otherwise, decreases the product. Repetition leaves just $v, 1-v$ as the nonzero weights, since $v > 1/2$. As $1/[v(1-v)] < (46/15)^2$, the determinant argument at $q = 1$ gives $\mathbb{P}\{Y > u\} \le (46/15)e^{-u/2}$. The first-moment and tail integration in Lemma S1.11, with cutoff $A = 13/2$ and scale $s = 2$, has tail integral $(46/15)se^{-A/s} \le 5/21$. Thus

$$\frac{\mathbb{E}Y^p}{p!} \le u_p := \frac{16}{21}\frac{A^{p-1}}{p!} + \frac{5}{21}\sum_{j=0}^{p-1}\frac{A^j s^{p-1-j}}{j!}.$$

Direct rational arithmetic gives $u_2 = 9/2$, $u_3 < 16$, and $u_4 < 64$. For $p \ge 4$, the two components grow by factors at most $A/(p+1) < 4$ and $s + A/p \le 29/8 < 4$, respectively. This proves all required bounds in this case.

For the remaining spectra, set

$$P = R^2/\eta^2, \quad T = X^TPX, \quad \bar r = \operatorname{tr}(RP), \quad C = R + P(\bar r I - 2R), \quad Q = X^TCX.$$

Lemma S1.8 gives $C \succeq 0$, $\operatorname{tr} C = \operatorname{tr} R - \bar r$, and $\Delta_+ \le T + \Delta_+ Q$. Write ρ_i for the eigenvalues of R/η and $a = \sum_i \rho_i^3$; the eigenvalues of C/η are $c_i = \rho_i + a\rho_i^2 - 2\rho_i^3$.

We first record a scalar monotonicity fact. Put $\mathcal{J}(z) = -\log(1-z) - z$ and $z = (12/5)(x + Ax^2 - 2x^3)$. Then

$$x \longmapsto \frac{\mathcal{J}((12/5)(x+Ax^2-2x^3))}{x^2} \quad \text{is increasing on } (0, 7/20], \qquad 1/2 \le A \le 1. \tag{S1.44}$$

Here $0 < z < 1$: the defining cubic is increasing in x and A on the stated rectangle, and its maximum gives $z < 1$. To verify the derivative sign using the existing rational logarithm bound $-\log(1-z) \le \mathcal{N}(z)/\mathcal{D}(z)$, it suffices that

$$xz'(x)z\mathcal{D}(z) - 2\big(\mathcal{N}(z) - z\mathcal{D}(z)\big)(1-z) \ge 0.$$

After division by x^3, this is a polynomial in (A, x). Its Bernstein coefficients on $[1/2, 1]\times[0, 7/20]$ are all greater than $49/500$, by the coefficient rule already given above. This is a finite rational certificate of (S1.44).

Use the following bounds in the three indicated spectral intervals:

interval for κ	r	A	q_*	c_*	d
$[15/16, 47/50]$	$87/250$	$1087991/1250000$	$13/250$	$37/100$	$31239/10000$
$[47/50, 19/20]$	$171/500$	$111/125$	$1/20$	$183/500$	$313/100$
$[19/20, 1]$	$5/16$	1	$1/25$	$7/20$	$63/20$

In every row, the largest eigenvalue of R/η contributes at most q_* to C/η; the other eigenvalues satisfy $\rho_i \le r$, $\sum_{i\ne\text{top}} \rho_i^2 \le r^2$, and $c_i \le \rho_i + A\rho_i^2 - 2\rho_i^3$. Moreover $r + Ar^2 - 2r^3 < c_*$ and $b - \operatorname{tr} C \ge d\eta$.

For clarity, these bounds follow directly from the endpoint spectra. On $[\kappa_{\mathrm{lo}}, \kappa_{\mathrm{hi}}]$ in the first two rows put $\varepsilon = 1 - \kappa_{\mathrm{lo}}^2$. The displayed r exceeds $\sqrt{\varepsilon}$. Since $z^3 + (1-z^2)^{3/2}$ increases for $z \ge 1/\sqrt{2}$,

$$a \le \kappa_{\mathrm{hi}}^3 + (1 - \kappa_{\mathrm{hi}}^2)^{3/2} \le A.$$

For the first row use $\sqrt{1-(47/50)^2} < 171/500$; for the second use $\sqrt{1-(19/20)^2} < 5/16$. The largest contribution is at most $\kappa_{\mathrm{hi}}\varepsilon^2 + \kappa_{\mathrm{hi}}^2 \varepsilon r < q_*$. For the last row it is at most $\varepsilon^2 + (5/16)\varepsilon < 1/25$, where $\varepsilon \le 39/400$. Finally $a \ge \kappa^3$ gives the gap from $23/10 + \kappa_{\mathrm{lo}}^3 > d$ in each row. All stated comparisons are rational.

Fix $\theta_0 = 6/5$. Gaussian comparison and (S1.44) now give

$$\begin{aligned} \log \mathbb{E} \exp\left(\frac{\theta_0 (Q - \operatorname{tr} C)}{\eta}\right) &\le \frac{1}{2} \sum_i \mathcal{J}(2\theta_0 c_i) \\ &\le \frac{1}{2}\big[\mathcal{J}(2\theta_0 q_*) + \mathcal{J}(2\theta_0 c_*)\big]. \end{aligned}$$

Indeed, the sum over the remaining eigenvalues is at most $\sum_{i \ne \mathrm{top}} (\rho_i/r)^2 \mathcal{J}(2\theta_0(r + Ar^2 - 2r^3))$. The following scalar bounds therefore imply $\mathbb{P}\{Q > b\} \le p_0$ and $(177/50)e^{-A_0/s_0} \le p_0$, where $s_0 = 50/23$:

row	ℓ_{top}	ℓ_{rest}	p_0	A_0
1	67/500	219/100	91/2000	947/100
2	16/125	527/250	7/160	239/25
3	101/1000	1833/1000	19/500	10

Specifically, $\ell_{\mathrm{top}} > -\log(1 - 2\theta_0 q_*)$ and $\ell_{\mathrm{rest}} > -\log(1 - 2\theta_0 c_*)$. With $F_* = (\ell_{\mathrm{top}} + \ell_{\mathrm{rest}} - 2\theta_0(q_* + c_*))/2$, we have $e^{F_* - \theta_0 d} < p_0$. The tail family (S1.29), with $q = 23/25$, also gives $\mathbb{P}\{Y > u\} \le (177/50)e^{-u/s_0}$, since $\sqrt{25/2} < 177/50$. Let $E = \{Q > b\}$. On E^c, the fixed comparison implies $Y \le (1-b)T/(1-Q) \le T$. Integrating the survival bound p_0 below A_0 and $p_0 e^{-(u - A_0)/s_0}$ above it gives

$$\mathbb{E}[h(Y)\mathbf{1}_E] \le p_0 \mathbb{E} h(A_0 + Z) \tag{S1.45}$$

for every nonnegative increasing differentiable h with $h(0) = 0$, where Z is exponential with mean s_0. No independence of E and Y is used.

We also retain spectral information in the Gaussian comparison for T. Put $g_p = 2^{-p}\binom{2p}{p}$. If G_i are independent standard Gaussians and p_i are the eigenvalues of P, Jensen's inequality gives $\mathbb{E}(\sum_i p_i G_i^2)^p/p! \le g_p$ for every integer $p \ge 1$. Its second coefficient is exactly $1/2 + \sum_i p_i^2$. On each of the first two intervals,

$$\sum_i p_i^2 \le \kappa^4 + (1 - \kappa^2)^2 \le \zeta_* := \kappa_{\mathrm{hi}}^4 + (1 - \kappa_{\mathrm{hi}}^2)^2.$$

For the third row put $\zeta_* = 1$. Gaussian comparison therefore bounds the full MGF of T by

$$1 + \lambda + (1/2 + \zeta_*)\lambda^2 + \sum_{p \ge 3} g_p \lambda^p.$$

These coefficients belong to the explicit Gaussian majorant.

Now use $h_\lambda(x) = e^{\lambda x} - 1 - \lambda x$ in (S1.45). The facts $\mathbb{E}Y \le 1$, $\mathbb{E}T = 1$, and $Y \le T$ on E^c give

$$\mathbb{E} e^{\lambda Y} \le 1 + \lambda + \mathbb{E} h_\lambda(T) + p_0 \mathbb{E} h_\lambda(A_0 + Z).$$

For $0 < \lambda < 1/4$ all terms are finite. The explicit majorant has coefficients

$$u_2 = \frac{1}{2} + \zeta_* + p_0 S_2, \qquad u_p = g_p + p_0 S_p \ (p \ge 3), \qquad S_p = \sum_{j=0}^{p} \frac{A_0^j s_0^{p-j}}{j!}.$$

In every row direct rational arithmetic gives $u_2 < 9/2$ and $u_p < 4^{p-1}$ for $p = 3, 4, 5$. For $p \geq 5$, $g_{p+1}/g_p = (2p+1)/(p+1) < 2$, while

$$S_{p+1} = s_0 S_p + \frac{A_0^{p+1}}{(p+1)!} \leq \left(s_0 + \frac{A_0}{p+1}\right) S_p \leq \left(\frac{50}{23} + \frac{10}{6}\right) S_p < 4S_p.$$

This proves all the required coefficient bounds.

For completeness, all exponential comparisons have finite rational certificates. The positive Taylor polynomial of degree 24 proves $e^{13/4} > 644/25$ and, in each row,

$$e^{\ell_{\mathrm{top}}} > \frac{1}{1 - 2\theta_0 q_*}, \qquad e^{\ell_{\mathrm{rest}}} > \frac{1}{1 - 2\theta_0 c_*}, \qquad e^{\theta_0 d - F_*} > \frac{1}{p_0}, \qquad e^{A_0/s_0} > \frac{177}{50 p_0}.$$

The accompanying checker verifies these and all stated rational comparisons. □

Lemma S1.13 (General upper exponential moment)**.** *Let $K \geq 1$ and $w \geq 23/10$. If $K^2 \operatorname{tr} R + wK^2 L\eta \leq b < 1$ and $Y = (1-b)\Delta_+$ has mean at most one, then* (S1.27) *holds with $\nu_+ = (9/2)K^2 L$.*

Proof. Put $M = K^2 L$ and $d = L - 1 \geq 0$. Lemma S1.11 already gives the required bounds for every moment of order $p \geq 3$. It remains to prove

$$\frac{\mathbb{E}Y^2}{2M} - \frac{1}{2M} \leq \frac{9}{2}. \tag{S1.46}$$

Indeed these moment bounds give $\mathbb{E}e^{\lambda Y} \leq 1 + \lambda/(1-\nu_+\lambda) + \lambda^2/2 \leq \exp[\lambda/(1-\nu_+\lambda)]$.

First suppose $\kappa \leq 15/16$. The capped-weight product estimate in Lemma S1.12 is deterministic, so it gives $\mathbb{P}\{Y > u\} \leq (46/15)e^{-u/(2K^2)}$ for every K. Use cutoff $(13/2)K^2 L$ in (S1.43). Its tail integral is at most $(5/21)e^{-5d/4}$, and hence

$$\frac{\mathbb{E}Y^2}{2M} - \frac{1}{2M} \leq \frac{13}{4} + \frac{5}{21} e^{-5d/4}\left(\frac{13}{4} + \frac{2}{L}\right) - \frac{e^{-2d}}{2L} \leq 4.$$

For the last inequality, differentiate as in the proof of Lemma S1.11. After multiplication by $L^2 e^{2d}$, the negative derivative is bounded below by

$$\frac{5}{21}\left(\frac{65}{16}L^2 + \frac{5}{2}L + 2\right) - L - \frac{1}{2},$$

whose coefficients in $L - 1$ are positive. The value at $L = 1$ is 4.

For $\kappa > 15/16$, retain the three spectral rows and their p_0, A_0, ζ_* from Lemma S1.12. Those bounds for the eigenvalues of C/η do not depend on K. Gaussian comparison, now centered at $K^2 \operatorname{tr} C$ and using parameter $\theta_0/(K^2\eta)$, gives

$$\mathbb{P}\{X^T C X > b\} \leq p_0 e^{-(69/25)d}.$$

Indeed $b - K^2 \operatorname{tr} C \geq K^2\eta[(23/10)L + a]$, so the earlier exponent gains $(6/5)(23/10)d = 69d/25$. The generic tail with $q = 23/25$ matches this probability at threshold $K^2(A_0 + 6d)$, because $(50/23)(69/25) = 6$. The same rare-event integration, applied only to the second moment, therefore gives

$$\frac{\mathbb{E}Y^2}{2M} - \frac{1}{2M} \leq \frac{m_4}{K^2 L}\frac{1 + 2\zeta_*}{6} + \mathcal{R}(d) - \frac{e^{-2d}}{2L}, \tag{S1.47}$$

where m_4 is any uniform upper bound on the fourth moments of the unit directional marginals, and

$$\mathcal{R}(d) = \frac{p_0 e^{-19d/25}}{1+d}\left[\frac{(A_0 + 6d)^2}{2} + \frac{50}{23}(A_0 + 6d) + \left(\frac{50}{23}\right)^2\right].$$

Here Gaussian averaging gives $\mathbb{E}T^2 \le m_4(1+2\zeta_*)/3$; on the complementary event $Y \le T$. No independence between that event and Y is needed. The function $\mathcal{R}$ decreases. To check this directly, write its polynomial factor as $P_0 + P_1 d + P_2 d^2$, where

$$P_0 = A_0^2/2 + (50/23)A_0 + (50/23)^2, \quad P_1 = 6(A_0 + 50/23), \quad P_2 = 18.$$

With $c = 19/25$, the derivative numerator has coefficients $P_1 - (1+c)P_0$, $2P_2 - c(P_0 + P_1)$, $P_2 - c(P_1 + P_2)$ and $-cP_2$; all are negative in every row.

We specify the remaining finite scalar certificates. For a variance-one marginal Z, moment interpolation gives, for every $x > 0$,

$$m_4 \log\left(1 + \frac{\cosh x - 1}{m_4}\right) \le \frac{K^2 x^2}{2}, \qquad m_4 = \mathbb{E}Z^4.$$

The left side increases strictly with m_4. Thus any $m_0, E, x > 0$ with $E \ge K^2$ and

$$m_0 \log\left(1 + \frac{\cosh x - 1}{m_0}\right) > \frac{Ex^2}{2} \tag{S1.48}$$

proves $m_4 < m_0$. For $1 \le L \le 51/50$, take $m_0 = 4$, $x = 5/2$ and $E = e^{1/25}$. The difference in (S1.48) exceeds $1/100$. Substitute $m_4 = 4$, $\mathcal{R}(d) \le \mathcal{R}(0)$ and $e^{-2d}/L \ge e^{-1/25}/(51/50)$ into (S1.47); the result is less than $9/2$ in each row.

For the remaining bounded range, use the 334 intervals

$$y \in [j/1000, (j+1)/1000], \qquad 376 \le j < 710, \qquad L = 1 + y^4.$$

Their union overlaps both $L = 51/50$ and $L = 5/4$. On a cell put $y_- = j/1000$, $y_+ = (j+1)/1000$, $d_\pm = y_\pm^4$, $L_\pm = 1 + d_\pm$, and

$$c(y) = 3 + \frac{9}{2}y^2 - \frac{21}{10}y^4 - \frac{4}{5}y^6.$$

Let E_+ be an upper bound for e^{2d_+} rounded upwards to a multiple of 10^{-12}, and similarly round $m_+ = c(y_+)E_+L_+$ upwards. Choose x by rounding $y_+(29/5 + 3y_+^2)$ to the nearest multiple of $1/20$, with ties rounded upwards. For every cell, the difference in (S1.48), with $(m_0, E) = (m_+, E_+)$, exceeds $1/10000$. Thus $m_4 < m_+$ throughout that cell. Using the monotonicity of $\mathcal{R}$, an upper bound for (S1.47) is

$$\frac{m_+ e^{-2d_-}(1+2\zeta_*)}{6L_-} + \mathcal{R}(d_-) - \frac{e^{-2d_+}}{2L_+}.$$

Each of the $3 \cdot 334$ rationally bounded values is less than $9/2$, with margin greater than $1/125$. For $L \ge 5/4$, Lemma S1.11 gives the required bound directly. This covers every $L \ge 1$ and proves (S1.46).

For these checks use (S1.6)–(S1.7) with exponential Taylor degree 48 and 24 logarithmic terms, reducing logarithm arguments by powers of two. Only logarithmic lower bounds are needed. Round outwards to multiples of 10^{-12} throughout. □

The two mean lemmas and the two exponential-moment lemmas prove the one-row assertions stated at the start of this subsection. Applying Lemma S1.2 gives

$$(\gamma, \delta) = (47/20, 3/2) \quad (K \ge 1), \qquad (\gamma, \delta) = (58/25, \sqrt{2}) \quad (K = 1).$$

Together with the lower-tail estimates, this proves Theorem S1.1.

S2 Lean formalisation and AI assistance

This section records the scope of the formal proof, the external numerical checks, and the roles of the author and the tools. The disclosure follows the recommendations of the *Leiden Declaration on Artificial Intelligence and Mathematics*.[1]

Both numerical quartets in Theorem S1.1 have completed *hybrid verification*: Lean checks the analytic implication from explicit finite arithmetic conditions, and exact Python programs check those conditions externally. Section S2.3 explains what remains outside Lean's kernel.

S2.1 The formal statement and its scope

The formal statement uses an arbitrary probability space and measurable random rows, with $m \geq n \geq 1$. Its probabilistic assumptions are those of Theorem S1.1: independent rows, centring, isotropy and the directional sub-Gaussian MGF bound. Row distributions may differ, and coordinates within a row may be dependent. Measurability and integrability are explicit in the definitions: directional first and second moments, and every directional exponential moment, have ordinary finite expectations. These fields express the usual meaning of the assumptions; they add no independence or higher-moment hypothesis.

The conclusion gives both non-strict tail events in (S1.1)–(S1.2) for every $t \geq 0$. Singular values are the square roots of all n eigenvalues of the Gram matrix, including zeros. Thus the result includes rank deficiency, $n = 1$, $t = 0$ and nonpositive lower thresholds. The constants are represented exactly as

$$(96/25, 79/50, 47/20, 3/2) \quad (K \geq 1), \qquad (33/25, 28/25, 58/25, \sqrt{2}) \quad (K = 1).$$

The first quartet remains valid at $K = 1$, where the second is sharper.

In namespace `SingularValueBounds.KL`, the file `KL/Theorem.lean` exposes:

```
singular_value_tails_general_of_certificates
singular_value_tails_one_of_certificates
theorem_explicit_constants_of_certificates
```

The first two results give the general and $K = 1$ bounds; the third combines them. Their additional hypothesis is `FinalCertificateConditions`, defined in `KL/Certificates/Final.lean`. It contains only the finite numerical conditions described in Section S2.3. No one-row estimate, deterministic barrier budget, measurability assertion or analytic inequality remains as an extra hypothesis.

S2.2 Correspondence with the written proof

The formalisation follows Section S1, including the barrier arguments imported from the main paper. Its principal groups are listed below; the source documentation gives the detailed correspondence to LaTeX labels.

1. *Barriers and transfer.* The `Barrier/` modules and `Transfer.lean` construct the exact inverse-trace-preserving shifts, their measurable adaptive process and the least admissible outward corrections. They prove positive definiteness, preservation of the resolvent trace at exact row updates, the accumulated correction budgets and the final Gram-matrix comparison. The probabilistic iteration proves independence of each new row from the entire measurable preceding history, together with exponential integrability and the MGF bounds. Chernoff optimisation gives Lemma S1.2 and its mixed-cap application.

[1] `https://leidendeclaration.ai/`

2. *Gaussian, entropy and numerical tools.* The `Weighted/` modules prove the Gaussian determinant comparison, directional moment averaging, weighted entropy and the integration arguments for the upper mean. The `Certificates/` machinery and rounded-arithmetic modules in `Upper/` prove how exponential and logarithmic enclosures, outward rounding, Bernstein coefficient bounds and finite domain coverage imply the required analytic inequalities. These are general Lean proofs, not conclusions drawn from numerical samples.

3. *The lower tail.* The `Lower/` modules prove the fourth-moment and lower Laplace bounds, the deterministic exact-shift comparisons, the general spectral profile, and the covariance and rank-one refinements at $K = 1$. The `Barrier/Mixed` family proves the measurable mixed-cap process and its total pathwise cost. Lean checks the actual profiles, their regularity and the strict scalar cost bounds yielding $\alpha = 3.84$ and $\alpha = 1.32$. Scaling is restored in the comparison matrices and one-row estimates; homogeneity of the nonlinear exact shift is not assumed.

4. *The upper tail and assembly.* The `Upper/` modules prove the exact shift events, Gaussian and joint tails, extended rates, the coefficient-two comparison matrix, determinant bounds, both upper means, and the tail-to-moment and MGF arguments. They establish the required integrability and apply the finite numerical conditions through the proved analytic rules. Finally, `Theorem.lean` supplies the lower and upper row estimates and barrier budgets to the transfer theorem, obtaining both quartets from the original row assumptions and the finite certificate bundle.

Some formal statements use the generality needed here. Weighted entropy has finite-integrability hypotheses, which are proved at each application; the most general extended-real formulation is not a separate formal result. The mixed-cap budget is proved for continuous profiles satisfying the stated monotonicity, positivity and integral conditions, covering both profiles used here. A clipped inverse, a trace potential and a Fubini/Young inverse-integral identity justify the calculation without differentiating a profile or its inverse at a breakpoint. Lean also proves the conversion of the singular profile integral to its nonsingular angular form and the one-dimensional case. These formulations justify the same estimates as the written proof.

S2.3 Finite certificates and the verification boundary

The eleven fields of `FinalCertificateConditions` are grouped below. The names in the right column are the actual Lean field names.

Finite conditions	Fields
General lower profile and plateau	`generalLower`
Endpoint lower table and far regime	`endpointLowerTable`, `endpointLowerFar`
Upper majorants, norm determinants and extended rates	`majorant`, `normDeterminant`, `rate`
General upper-mean cells and entropy	`generalMeanCells`, `generalMeanEntropy`
Endpoint upper mean	`endpointMean`
Upper exponential-moment comparisons	`endpointLogratio`, `generalSecond`

These fields consist of finite rational comparisons, coefficient identities, subdivision and partition conditions, and exact rounded-arithmetic transcripts. Lean proves that they imply inequalities throughout the required real domains, including equality boundaries and unbounded parameter regimes. Python checks the concrete finite data.

The original `check_main_constants.py` and the interface runners use exact integer and rational arithmetic. They cover the lower profiles, polynomial majorants and rates, upper means, entropy transcripts and exponential-moment comparisons. Two additional runners check elementary enclosure examples and repeat the profile-integral arithmetic; the latter comparisons are also proved directly in Lean and add no numerical premise. The original checker also checks

arithmetic in the main paper's sharpness examples, whose analytic proofs are outside the Lean theorem's scope. The completed proof map records the correspondence between fields and checkers.

The large finite calculations are not replayed in Lean's kernel. Python supplies no Lean proof terms, and no proof of the certificate bundle is inserted into the formal theorem. Exact arithmetic avoids floating-point rounding error. The combined verification nevertheless depends on the checker implementation, its execution, and the correspondence of its data and formulas to the Lean predicates. That correspondence was reviewed at source level. Recorded source hashes detect later changes to a reviewed transcription; they are not compiler-generated exports and do not prove correspondence. The numerical theorem therefore has hybrid verification, with a kernel-checked conditional analytic proof, rather than an unconditional proof entirely within Lean's kernel.

S2.4 Build, axiom audit and reproduction

The recorded environment uses Lean 4.19.0, Mathlib v4.19.0 and Python 3.14.0. Mathlib is pinned to the revision in `lake-manifest.json`. The complete verification passed on 20 October 2026, after retirement of the earlier standalone theorem. The library contained 147 `KL` modules and 32 shared supporting modules. Its audit covered 3,224 declarations: all 3,200 theorem declarations in the `KL` namespace, including generated equations and structure facts, and 24 selected shared supporting results. Lean's transitive axiom collector visited 53,478 constants. Only `propext`, `Classical.choice` and `Quot.sound` were reported, the standard logical principles used by Lean and Mathlib. There are no `sorry` proofs, added theorem-specific axioms or native-oracle proofs. Each of the three final public results also has a separate `#print axioms` report.

The audit checks that its targets belong to the checked kernel environment. Its report checker enforces the axiom list and target checkpoint, and checks that every current `KL` source is imported. An explicit theorem hypothesis does not appear in `#print axioms`: this audit checks the dependencies of the conditional analytic implication, while the external checks address its finite numerical hypothesis as explained above.

From the supplementary GitHub repository root, run the complete check with

```
./lean/check_hybrid.sh
```

It builds and audits the Lean library, runs the original arithmetic checker and all eleven interface runners. It rejects optimised Python execution, because the exact checkers use assertions. The command reuses the configured external workspace and pinned dependencies without silently installing or updating them.

The Lean README gives the setup; the completed plan gives the proof and certificate map; and the verification report and source manifest record the checked inventory and provenance of the historical run. The earlier standalone $n^{1/6}$ theorem has been retired, while its shared lemmas needed by the present proof remain in the library.

S2.5 AI assistance, responsibility and limitations

The authors of the accompanying paper carried out extensive mathematical work, yielding weaker bounds, before working with Codex. That research preceded, and is distinct from, the generation of this supplementary document. This entire document was generated by OpenAI Codex, identified by the authors as GPT-6, without direct input from its author.

Codex developed and refined the proof, searched the literature, wrote and ran the exact arithmetic checker, and prepared the document. It also developed the formalisation, resolved Lean elaboration errors, ran the checks and reviewed correspondence between the formal and

written arguments. The recorded formalisation work included parallel GPT-6.1 Sol Ultra workers and a coordinating agent. The models produced candidate arguments and code; Lean's kernel checked formal proof terms, and Python executed external finite arithmetic checks. Reviews by additional model instances do not constitute independent human review or remove the verification boundary in Section S2.3.

Marcus Webb takes full responsibility for the entire document: its mathematical arguments, exposition, computational certificates and attribution. This includes judging correspondence between the formal statement and the intended theorem, and between the Python checks and the formal numerical premises. The formalisation does not verify the main paper's applications or sharpness proofs, or establish priority, novelty or completeness of the literature assessment. Those remain matters of mathematical and scholarly judgment; successful compilation and exact arithmetic do not resolve them.

The supplementary materials, including this document and its LaTeX source, the Lean formalisation, exact arithmetic checkers and reproduction instructions, are available in the GitHub repository `https://github.com/marcusdavidwebb/SubGaussianTailBounds`.

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